A Structural Model of Reward Programs on Digital Platforms: The Case of Livestreaming

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Abstract

This paper presents a structural model to assess the effects of seller reward programs on digital platforms. The empirical context is a Korean livestreaming platform where sellers (streamers) broadcast content, earn revenue from viewer tips, and receive commission discount rewards through performance-based monthly tournaments. I collect streamer-time-level microdata on effort and revenue from streamer tracking websites and estimate a dynamic game to capture how reward program design affects forward-looking streamers' behavior and, in turn, platform revenue. Counterfactual simulations show that commission discount rewards motivate streamers—especially more profitable ones—to stream more and increase total revenue, but cause the platform to retain a substantially smaller share of that revenue. Because the latter effect quantitatively dominates, the current program is not optimal, and reducing the discount rate can improve platform revenue. These results highlight a central trade-off in platform design between incentivizing seller effort and preserving platform revenue.

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1 Introduction

To motivate their sellers, numerous digital platforms operate performance-based reward programs. For example, high-performing Uber drivers can earn cash rewards and gas subsidies through the Uber Pro program.[1] The eBay Top Rated Seller Program offers fee discounts and enhanced visibility for sellers based on their sales records and on-time shipment.[2]

What factors should a platform consider when designing such reward programs? I study this question in the context of the livestreaming industry. In this industry, sellers (called streamers) broadcast content such as video games and generate revenue by receiving tips from viewers.[3] Platform revenue is determined by three factors. First, total watch time, defined as the amount of time viewers spend watching streamers. Second, average profitability, i.e., how much tipping revenue streamers extract from a given amount of watch time. Third, the platform's share of the total revenue.

The reward program in the empirical context of this paper takes the form of a commission discount for popular streamers. This reward creates a trade-off among the three aforementioned factors. Offering the reward could raise platform revenue through two channels. First, streamers work more on average, leading to an increase in total watch time. Second, more profitable streamers, who earn higher tipping revenue per unit of watch time, could be disproportionately motivated by the commission discount reward. If such streamers occupy a larger portion of watch time, this increases the average tipping revenue per unit of watch time. However, the platform receives a smaller share of the generated revenue due to the commission discounts.

To empirically study this trade-off, I collect microdata from a leading livestreaming platform in South Korea. The environment is suitable for my study for two reasons. First, individual streamer-time data on effort and outcome measures are available, such as streaming hours, viewership, tipping revenue, and rewards program acceptance. Second, the benefits and requirements of the rewards program are straightforward. The key benefit is a commission discount, and acceptance is determined by monthly tournaments based on observable streamer performance metrics.

To assess how different reward program designs affect heterogeneous streamers' effort choices and competition, I estimate a dynamic game model of streamers.

[1] https://www.uber.com/us/en/drive/uber-pro/, accessed on May 23, 2023

[2] https://www.ebay.com/sellercenter/protections/top-rated-program, accessed on May 23, 2023

[3] This paper does not incorporate ad revenue due to the lack of data. In my empirical context, financial reports from the focal platform show that about 80% of its revenue comes from tips.

The model is naturally dynamic, as it describes streamers' forward-looking decision-making: they stream more in the current period to improve their chances of receiving the commission discount and starting with a larger fan base in the future. Additionally, the game features two relevant interactions among streamers: (1) streamers with similar content compete for viewership, which means that a streamer's watch time may decrease if other streamers increase their streaming hours; (2) the reward program is determined through a performance-based tournament, so the likelihood that a streamer is accepted decreases as the number of other streamers who perform better increases.

In my model, streamers primarily differ along two dimensions: their profitability, measured by tipping revenue per unit of watch time, and the category of content that they broadcast. These differences reflect a streamer's personal characteristics, such as their entertainment skills and talent for creating specific types of content. Both dimensions affect platform revenue in the model. Whether more profitable streamers capture a larger share of watch time affects total tipping revenue, and the market-stealing effect from within-category competition influences total watch time.

To solve the model and estimate the parameters, two computational difficulties arise. First, the curse of dimensionality: streamers must account for the states and actions of thousands of potential competitors. I address this challenge using an oblivious equilibrium approach (Weintraub et al., 2008), in which streamers summarize the influence of competitors using a single state variable: the weighted sum of the streaming hours of other streamers broadcasting the same type of content.

Second, the standard logit-based approach for approximating watch time yields corner solutions for streaming effort choices, because marginal returns increase exponentially with effort. To address this limitation, I approximate watch time using a flexible function estimated with a tree-based model (XGBoost), which yields interior optimal effort choices and improves predictive performance.

Using the estimated model, I conduct three counterfactual simulations to evaluate the effects of different reward program design on platform revenue. First, I adjust the number of streamers who receive the reward in each period. Second, I change the magnitude of the commission discount provided as a reward. Third, I study a more granular program design, in which approval slots are allocated at the broadcasting category level instead of at the platform level.

Overall, the counterfactual results show that reducing the reward can improve platform revenue. As expected, offering commission discounts makes streamers stream

more, which increases total watch time (*watch time effect*). Additionally, it disproportionately motivates more profitable streamers, who can extract more tips per unit of watch time, thereby improving profitability measured by revenue per watch time (*profitability effect*).

However, the revenue improvement from these factors is not sufficient to offset the decrease in the platform's share of generated revenue due to the commission discounts (*platform share effect*). This situation is analogous to a scenario in which a firm's revenue does not increase despite offering price discounts due to inelastic demand.

In the first counterfactual simulation, I double and halve the number of streamers receiving the reward each period. When the number is doubled, both the watch time effect and profitability effect increase total revenue by 0.22% and 0.94%, respectively. However, the platform's share of total revenue decreases by 3.66%, resulting in a 2.53% decrease in platform revenue when all factors are taken into account. On the other hand, halving the number of streamers receiving the reward leads to a 3.03% increase in platform revenue compared to the original platform revenue.

Similar results emerge when modifying another design factor: the amount of commission discount. When the discount benefit increases to 15% (from the current 10%), the effects on watch time and profitability raise total revenue by 0.07% and 0.34%, respectively. However, due to a 6.63% decrease in the platform's share, the platform's revenue decreases by 6.24% overall. Conversely, reducing the commission discount benefit to 5% raises platform revenue by 6.08%.

I also conduct a third counterfactual simulation in which the number of streamers who can receive the reward is allocated at the category level (Game, Social, or Other), instead of at the platform level. The purpose of this exercise is twofold. First, it reveals whether and to what extent the platform can gain from adopting a more granular program design. Second, while the results above imply that reducing or even removing program benefits may improve platform revenue, such changes may be difficult to implement in practice.[4] Category-level reallocation could be an alternative way to increase platform revenue without unilaterally reducing overall streamer benefits.

Specifically, I increase the number of streamers receiving the discount reward in the Game category by two, while simultaneously reducing this number by two in the Social

[4]The streamers I interviewed all expressed very negative reactions to the idea of removing existing benefits. Similarly, Bewley (1998) found that, in practice, pay cuts are rarely implemented because managers believe they significantly harm employee morale, despite their justification in economic models with rational agents.

category. The key observation for this analysis is that since the current program offers rewards without differentiation among streamers in different categories, the marginal revenue improvement from providing a reward to each category is not equalized across categories. Therefore, giving more rewards to the more responsive category should increase platform revenue. I found that this design change improved platform revenue by 1.19%.

The main takeaway from the counterfactual simulations is that providing rewards that reduce a firm's revenue must be done carefully, taking into account sellers' responsiveness to the reward. As demonstrated in this study, such benefits can accumulate and significantly reduce the firm's revenue in the long-run equilibrium.

While this paper focuses on a specific institutional context, this takeaway is relevant for a wide range of platforms and industries. Because livestreaming platforms have similar business models across countries, the model and findings are applicable to other livestreaming platforms such as Twitch (US) and Huya (China).

More broadly, this takeaway applies to platforms that rely on commission-based revenue and use performance-based rewards to motivate sellers, including gig economy platforms (e.g., Uber, DoorDash) and online marketplaces (e.g., eBay, Etsy). In these settings, increasing seller rewards may stimulate effort and output but can also reduce platform revenue if the increase in seller activity does not sufficiently offset the reduction in the platform's commission share.

Related Literature This paper contributes to the literature on incentive schemes in a firm. Two strands of literature connect to this research question. First, from the perspective of personnel economics, the rewards programs can be considered as performance-based job promotions on a digital platform.[5] Such promotion incentives have been studied by Eriksson (1999), DeVaro and Morita (2013), and Belzil and Bognanno (2008), among many others. Second, salesforce management literature in marketing investigates how compensation plans, like bonuses and quotas, shape worker efforts. Examples include Misra and Nair (2011), Chung, Steenburgh, and Sudhir (2014), Daljord, Misra, and Nair (2016), Chung, Kim, and Park (2021).

The primary contribution of this paper is to develop a structural model that incorporates worker competition and relative performance-based compensation. In this paper's model, workers compete with each other. As a result, one worker's effort may negatively the affect other workers' outcomes. Furthermore, the model indirectly

[5]See Table 1 in Lazear (2018) for the taxonomy.

captures a feature wherein a worker has fewer chances to receive rewards when other workers perform better.

This paper is also related to the dynamic contests/tournaments design literature. A number of papers study this topic, particularly regarding how a designer should provide information for participants. Examples include Lemus and Marshall (2021), Bimpikis, Ehsani, and Mostagir (2019) and Mostagir, Chen, and Yeckehzaare (2019). While such literature mainly studies dynamics *within* a single tournament, I focus on a dynamic tournament that repeats over time by using a steady state equilibrium framework. This approach is suitable to capture dynamics *across* tournaments. For example, a participant of an online prediction tournament (e.g. Kaggle) may improve their coding skills and perform better in future tournaments.

Additionally, this paper contributes to a small but growing body of literature on the livestreaming industry. This new industry has provided a good environment for studying topics like congestion externality and influencer marketing. Some examples are Tudón (2021), Simonov, Ursu, and Zheng (2021), Huang and Morozov (2022), and Lu et al. (2021). In this strand of literature, this paper has two marginal contributions. First, it places its focus on the understudied role of streamer rewards programs. Second, it directly incorporates micro-level revenue information, enabling a more accurate capture of the marginal benefit of streamers' efforts.[6]

Lastly, from a methodological perspective, this paper relies on the literature of dynamic games in Industrial Organization, building upon the seminal work by Ericson and Pakes (1995). Solving a dynamic game poses a computational challenge called the curse of dimensionality.[7] I follow an idea developed in the oblivious equilibrium literature to address this challenge.[8] I additionally incorporate tournament between players within this framework to show its empirical usefulness for evaluating the rewards program designs.

[6]Obtaining such micro-level revenue information is often challenging due to the involvement of third parties. For instance, on platforms like Twitch, many streamers receive tips through third-party applications such as Twip and Toonation. As a result, even the platform itself does not have precise information about its streamers' revenue. In my empirical context, advertising such third-party applications is not allowed.

[7]For example, when there are 100 players and 10 possible states, a rational player has to account for 10^{100} possible configurations to develop a contingency plan.

[8]The key simplification is that each player solves a single-agent dynamic programming, believing that other players' states and actions are unaffected by the player's actions. This methodology describes markets well when market shares are not too highly concentrated (Weintraub, Benkard, and Van Roy (2008)). In my empirical context, even the top 1% of streamer-month observations take up 2.3% of the monthly market share as measured by watch time.

Roadmap Section 2 describes the general industry background, my specific empirical set up, and data sources. Section 3 presents my model. Section 4 describes the estimation procedure and model parameter estimates. Section 5 presents counterfactual simulations, and Section 6 concludes.

2 Empirical Context & Data

2.1 General Industry Background

A livestreaming platform is an online space where sellers, called streamers, broadcast various types of content, such as singing or playing video games. Each streamer functions like a television channel, as shown in Figure 1.

The livestreaming industry has become a flourishing business over the last decade, both in terms of time and money spent. Twitch, a dominant firm in the U.S. market, had 31 million average daily visitors and 8 million monthly active streamers in 2021.[9] This popularity has made streaming a lucrative venture for some. For example, a top streamer on Twitch is estimated to earn about $2 million per year from streaming on the platform.[10] The majority of users are young, making the industry appear promising for further growth. For example, as of 2022, 73% of Twitch viewers were under the age of 35.[11]

Livestreaming platforms are not only popular in the United States but also in other countries. In China, the leading platforms include Douyu, Huya, Douyin, and Bilibili, followed by Zhanqi and LiveQQ, all with massive user bases. Douyu alone had 163.6 million monthly active users in 2019.[12] In Japan, YouTube Live and Twitch are popular, competing with local rivals such as Niconico. In South Korea, during the data period of this paper, the top two platforms were AfreecaTV and Twitch, while YouTube Live also holds a significant market share.[13]

Business Model Streamers and livestreaming platforms primarily generate revenue from viewer tipping. Viewers donate money to their favorite streamers, similar

[9] https://twitchadvertising.tv/audience/, accessed on May 1, 2022.
[10] https://www.cashnetusa.com/blog/highest-paid-twitch-streamers-world/
[11] https://www.businessofapps.com/data/twitch-statistics/, accessed on August 29, 2022.
[12] https://old.capitalwatch.com/article-4697-1.html
[13] As of 2025, Twitch no longer operates in the Korean market, and AfreecaTV has been rebranded as SOOP. In place of Twitch, Naver (the dominant search engine in Korea) launched its own streaming service, Chzzk, thereby preserving the duopoly structure of the livestreaming market.

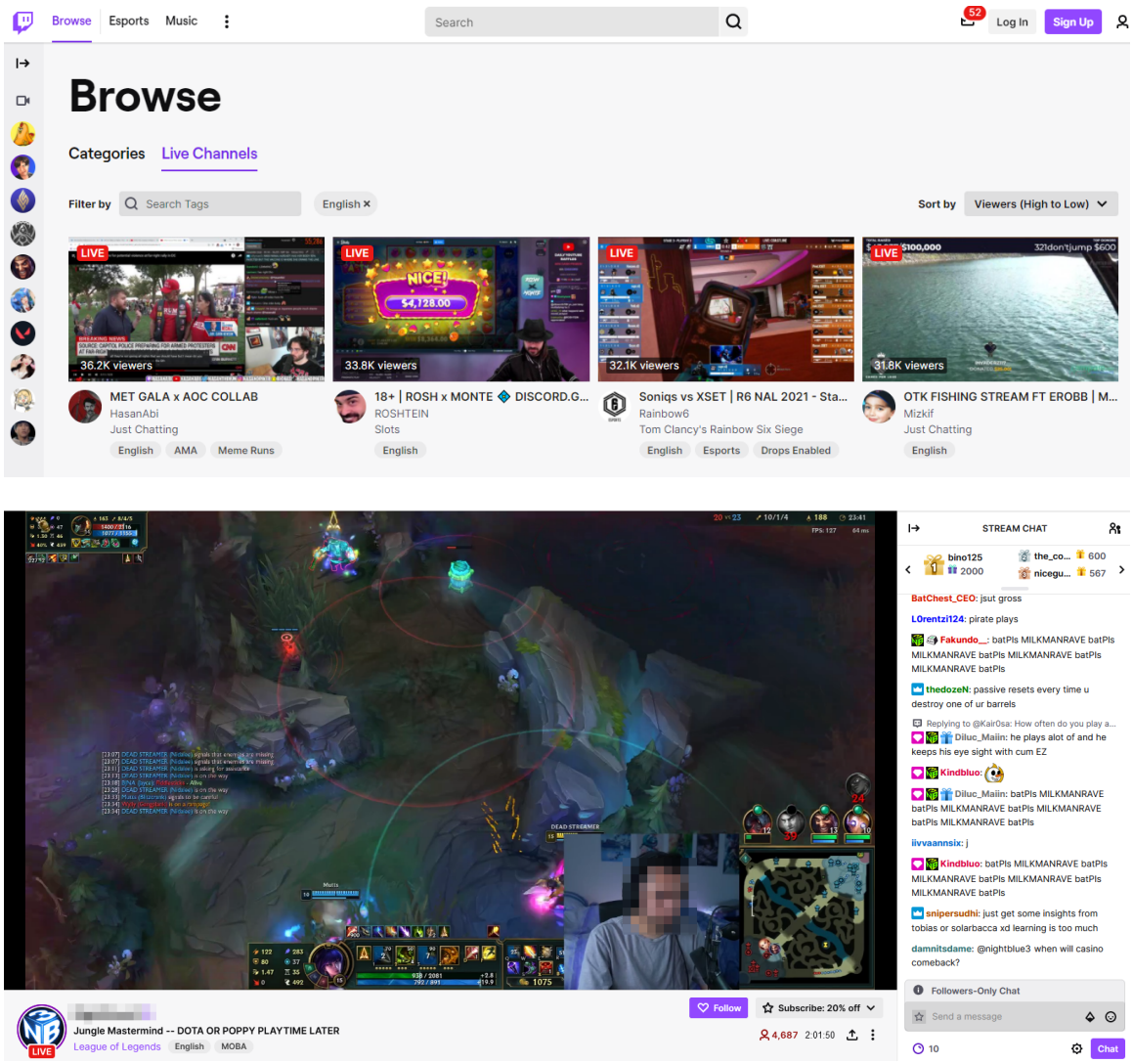


Figure 1. Twitch Homepage (Top) and Streamer Channel (Bottom)

Notes: Twitch is the dominant live-streaming platform in the US. On the right is the chatroom where viewers interact with the streamer. The interface shown here is typical across different streaming platforms, such as YouTube Live and AfreecaTV (currently SOOP)

to tipping street performers. The platform earns revenue by charging a commission on tipping income.

In general, viewers can watch streamers' content without paying, but many appear to derive utility from tipping because it facilitates communication with streamers they like. Tips are typically accompanied by messages of encouragement or questions, and streamers respond to these messages. Viewers donate to streamers they like for two common reasons. First, viewers want to financially support streamers so that they can keep producing content. Second, viewers enjoy responses from streamers—similar to a paid opportunity to interact with a celebrity they like—as well as recognition from other viewers.

Although other revenue sources, such as advertising, exist, their role is limited, especially in the empirical context of this paper. Fact sheets from AfreecaTV (the focal platform of this paper) show that in 2019, 72% of the company's revenue came from tipping, while advertising accounted for only 17%.[14] In this paper, I assume that all revenue comes from tipping.

Rewards programs are commonly used to motivate streamers in this industry. For example, Twitch operates two such programs: Affiliate and Partner. Facebook Gaming classifies its streamers into three tiers: general gaming creators (default), Level Up creators, and Partners. The requirements and benefits of these programs vary across platforms. In general, platforms evaluate streamers based on total streaming time, viewership, and the number of "fans" (i.e., viewers who have bookmarked the streamer).[15]

2.2 AfreecaTV and Its Rewards Program

The empirical context of this study is AfreecaTV, a leading livestreaming platform in South Korea.[16] Established in 2006, AfreecaTV has been a market leader in South Korea, both in terms of market share and innovation. For example, the platform introduced a tipping system for viewers to support streamers as early as 2007, even before

[14]Source: https://corp.sooplive.co.kr/ir.php?page=irbbs, 2020 earnings release, retrieved on February 12, 2026.

[15]For example, the minimum requirements to qualify as a Twitch Affiliate are: $\geq$ 50 fans, $\geq$ 500 minutes of total streaming time, $\geq$ 7 broadcast days, and $\geq$ 3 average viewers over the past 30 days. Source: https://help.twitch.tv/s/article/joining-the-affiliate-program, retrieved on March 24, 2022.

[16]The platform name officially stands for "Anybody can FREEly broadCast TV," but a more commonly cited explanation is that the company had a slogan, "New Continent for Individual Broadcasting," and mistakenly interpreted Africa, rather than America, as a new continent.

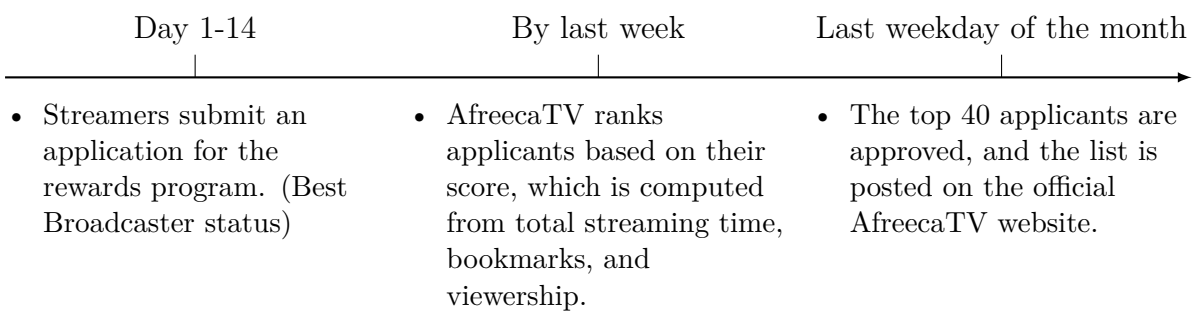


Figure 2. Timeline of the rewards program on AfreecaTV for each month

Notes: The timeline applies to the data period (from Oct 2019 to April 2020). The score calculation table is available in Table A.7 in the appendix.

Twitch, the dominant player in the US market, was launched in 2011. AfreecaTV has maintained a mutually beneficial relationship with the popular e-sports culture in South Korea, regularly broadcasting high-quality video game competitions and hosting numerous streamers focused on video game content.

One prominent goal for streamers on AfreecaTV is to achieve *Best Broadcaster* status, which is the rewards program that this paper explicitly focuses on.[17] The primary benefit for streamers of achieving this status is a commission discount. Prior to obtaining this status, streamers on AfreecaTV receive only 60% of the revenue generated from viewer tips, while the platform takes a 40% commission. However, the commission rate for a Best Broadcaster is reduced by 10 percentage points, allowing streamers to receive 70% of the revenue they generate. According to the collected data, this benefit results in about a 988 USD increase at the streamer-month level for streamers who newly receive the status.[18]

Importantly, the reward is effectively *permanent*. AfreecaTV approves new streamers into its rewards program on a monthly basis, as shown in Figure 2. Once approved, the only requirement to maintain eligibility is to stream for a minimum of five days, totaling 15 hours per month. This condition is easily achievable for streamers who have gained enough popularity to be approved for the rewards program. As a result, while only 40 out of around 6,000 streamers get the reward each month, streamers

[17]Strictly speaking, there is one additional rewards program called the Partner Broadcaster. However, this rewards program involves a private contract between the platform and a streamer, and I can only observe the timing. Furthermore, acceptance to this rewards program happened only 14 times during the data period. For the purposes of this paper, I assume that the benefits for partner-level streamers are the same as those for Best-level streamers.

[18]To be precise, the original unit of revenue is an item called a starballon. Throughout the paper, I use the approximation that 10 starballons are equivalent to 1,000 KRW or 1 USD.

Table 1. Variable Definitions

Variable	Description
streaming hour	The number of hours a streamer streams each month.
watch time	The total duration that viewers spend watching a streamer. For example, if two viewers watch a streamer for 10 and 5 hours in a month, the streamer's watch time is 15 hours.
(total) revenue	The amount of "tip" that a streamer receives, which is split linearly between the streamer and the platform.
(broadcasting) category	A category for a streamer's broadcasting content (e.g. Game, Social)
bookmark	The number of viewers who have bookmarked a streamer, which serves as a measure of the streamer's fan base.
score	A performance metric used to determine eligibility for the rewards program. It is computed based on viewership, bookmarks, and cumulative streaming time.
$\mathbb{1}\{Best\ Broadcaster\}$	A dummy variable that indicates whether a streamer has been approved for the rewards program or not.
$\mathbb{1}\{exit\}$	A dummy variable that indicates whether a streamer appears in the next period or not.

who received the reward account about 23% of the observation (See Table 2).

2.3 Data

I collect the main dataset from two streamer tracking websites and AfreecaTV's official webpage. Specifically, I gather variables at the individual streamer-day level, including revenue, streaming hours, bookmarks, viewership, and watch time. Additionally, I assemble streaming hours by (broadcasting) category at the streamer-month level, which provides information about the content on which streamers focus. Lastly, information on the reward program's benefits and requirements, as well as monthly announcements of reward recipients, is obtained from the platform's official website. Table 1 provides definitions of the key variables used in this paper. Appendix A.1 presents details on dataset construction.

I aggregate the data at the individual streamer-month level for two reasons. First,

Table 2. Descriptive Statistics of Key Variables

Variable	Mean	Std	P10	P50	P90
streaming hour	82.94	91.77	0.00	57.00	204.00
watch time (1,000)	10.63	118.02	0.00	0.77	10.22
(total) revenue (1,000$)	3.63	9.58	0.00	0.38	10.03
streamer's revenue (1,000$)	2.40	6.49	0.00	0.24	6.52
streamer score	44.36	27.33	12.00	38.40	81.60
$log(bookmark)$	6.97	2.11	4.45	6.76	9.93
$\mathbb{1}\{BestBroadcaster\}$	0.23	0.42	0.00	0.00	1.00
$\mathbb{1}\{exit\}$	0.08	0.27	0.00	0.00	0.00

Notes: The unit of observation is an individual streamer-month. The sample covers October 2019–April 2020 and includes 8,282 unique streamers and 41,221 observations. The streamer score and $log(bookmark)$ are measured at the beginning of each month. The score is computed using the platform's official score conversion table based on observed total streaming hours, bookmarks, and viewership.

this approach substantially simplifies model building, given that approval for the rewards program occurs on a monthly basis. Second, aggregation smooths out day-of-the-week effects.

Table 2 presents descriptive statistics. Two observations are worth noting. First, as expected, watch time and tipping revenue are highly skewed. Streamers at the median earn only about 236 USD per month, while the top 10% earn more than 6,000 USD per month. Second, streamers who have obtained the reward (Best Broadcaster status) account for a substantial number of observations. While only 40 streamers receive this status each month, because the reward is effectively permanent, recipients become common on the platform.[19]

2.4 Empirical Observations

2.4.1 Streamers Focus on One Broadcasting Category

A streamer may broadcast different types of content on different days. For instance, a streamer may stream a video game such as League of Legends one day and host a

[19]The data period ranges from October 2019 to April 2020. While some data are available after April 2020, I focus on this period because of two events that significantly affected the industry: COVID-19 and the increase in monthly program slots at the end of April. Starting in April 2020, the state of the industry appears to be non-stationary. For instance, the fraction of streamers accepted into the rewards program has consistently increased over time, making it difficult to fit the data into a stationary model.

song contest the next.

However, streaming hour data at the individual streamer–month–category level reveal that the majority of streamers adopt a "focus strategy." According to my category definition (Game, Social, or Other), 90% of streamers have one category that accounts for more than two-thirds of their total streaming hours.

Therefore, in my model, I assume that streamers have one broadcasting category that does not change over time. This category matters because streamers in different categories may have different marginal revenues or costs of streaming time, and streamers compete within a category. Streamers and their fans typically view streamers who broadcast similar content, such as the same video game, as their main competitors.

2.4.2 Profitability Varies Across Individuals

Profitability, measured by revenue per unit of watch time, varies substantially at the individual streamer–month level. The variable has a mean of 1.38 and a standard deviation of 14.22 USD per unit of watch time, indicating substantial dispersion across observations.

More than 80% of this variation is explained by individual streamer heterogeneity. This result is intuitive, since tipping revenue is heavily influenced by personal traits such as communication skills and appearance. Motivated by this evidence, the model assumes that each streamer has an individual-specific profitability parameter that is fixed over time.

Appendix A.4 presents fixed-effects regression results showing that the majority of profitability variation is explained by individual streamer fixed effects and that profitability is stable over time within individual streamers.

2.4.3 Not all eligible streamers receive the reward

One data limitation is that I can observe which streamers newly receive the reward, but I do not observe which streamers applied to the reward program and which did not. In the model, I need to determine two probabilities: the probability that streamers apply to the reward program ($\mathbb{P}(application)$), and the probability that streamers are approved for the reward program, conditional on application and score ($\mathbb{P}(approval \mid application)$).

To pin down $\mathbb{P}(application)$, I use the fact that some streamers have scores above the typical cutoff during the data period but do not receive the reward. Because

these streamers have sufficiently high scores to qualify, failure to receive the reward is informative about non-application. I therefore calibrate $\mathbb{P}(application)$ to match the fraction of such streamers who receive the reward in a given month.

In the model, streamers draw an i.i.d. shock in each period that determines whether they apply to the program. This shock governs the application decision, and conditional on application, the probability of receiving the reward depends on the streamer's score.

Two additional patterns are worth noting: (1) as expected, a streamer's probability of receiving the reward increases with their score; (2) despite the relatively low monthly application probability implied by this calibration, most eligible streamers eventually receive the reward in the long run.

Discussion of the limitations of the model is provided in Section **??**. Appendix A.5 provides details on the long-run probability that streamers eventually receive the reward, as well as an explanation for delays in application.

3 Model

The goal of the model is to capture how the rewards program affects heterogeneous streamers' decisions and, consequently, platform revenue. The model focuses on the dynamic problem faced by streamers in different states.

I use a dynamic game framework for two reasons. First, the rewards program is inherently dynamic: streamers exert effort today in anticipation of future benefits. Second, interactions across streamers matter. In each period, competition reduces individual revenue, and dynamically, the tournament-based approval process implies that other streamers' performance affects a focal streamer's likelihood of receiving the reward.

Streamer heterogeneity is captured by state variables. Streamers differ in content type and profitability (revenue per watch time), which are assumed to be time-invariant, as well as time-varying states such as bookmarks, scores, and Best Broadcaster (reward) status. In each period, streamers make dynamic decisions, taking into account both current payoffs and future expected benefits. In the following subsections, I describe the model and equilibrium more formally.

3.1 Streamer's Problem

3.1.1 Environment

Time Time is discrete and indexed by t. The time unit is a month, and the common discount factor is denoted by β.

Players The players in the game are streamers, indexed by j.

States Streamer j in period t is heterogeneous and represented by a state vector x_{jt}. This vector includes two time-invariant state variables: revenue per watch time α_j^{rev} and content category c_j. I use the subscript c to denote a category.

In addition, the vector includes three state variables that evolve over time: $\log(bookmark)_{jt}$, $score_{jt}$, and $r_{jt} \equiv \mathbb{1}\{Best\ Broadcaster\}_{jt}$, which indicates whether streamer j in month t has been approved for the rewards program.

Shocks Four independently and identically distributed shocks affect a streamer's decisions. (1) An exogenous exit shock, denoted by $\xi_{jt} \in \{0, 1\}$, determines whether a streamer exits the market. A streamer exits if $\xi_{jt} = 1$, capturing exits unrelated to profit, such as mandatory military service.[20] (2) A scrap value shock, denoted by $\phi_{jt} > 0$, captures a streamer's outside option. (3) A program interest shock, denoted by $\mathcal{I}_{jt} \in \{0, 1\}$, determines whether a streamer applies to the rewards program; a streamer applies if $\mathcal{I}_{jt} = 1$. (4) A streaming cost shock, denoted by $\gamma_{jt} > 0$, captures random factors that make streaming easier or harder, such as health issues or updates to video games.[21]

Actions In each month, a streamer makes two decisions: whether to exit and how many hours to stream. A streamer may exit either exogenously due to the shock ξ_{jt} or endogenously. Endogenous exit occurs if the scrap value draw exceeds the expected net present value of continuation. If the streamer does not exit, they choose their monthly streaming hours.

State Dynamics Three state variables, $\log(bookmark)_{jt}$, $score_{jt}$, and r_{jt}, evolve according to a Markov process. The first two states are discretized, and streamers

[20] This idea is similar to Chen and Xu (2022).

[21] For example, when a new update of a video game is released, streamers may find it easier to generate content by exploring new features.

can move up or down one interval or remain in the same interval in each period. The probability of moving to a higher interval increases with streaming effort.

For r_{jt}, I assume that streamers face approval probabilities $\mathbb{P}(r_{jt+1} = 1 \mid r_{jt} = 0)$ that depend on their score, conditional on $\mathcal{I}_{jt} = 1$ (i.e., they apply to the rewards program). If $\mathcal{I}_{jt} = 0$, the approval probability is zero. If $r_{jt} = 1$, then $r_{jt'} = 1$ for all $t' > t$, reflecting that approval is permanent.

Payoff Streamers derive quasi-linear utility from their share of revenue. This revenue depends on the watch time they generate in each period.[22] Streamers differ in their ability to monetize watch time, captured by their time-invariant revenue per unit of watch time, α_j^{rev}. Let $\mathcal{W}$ denote watch time, and let c_1, c_2 denote the parameters of the quadratic streaming cost. For streamer j in period t, who streams h_{jt} hours, the per-period payoff is

$$u_{jt} = \underbrace{(0.6 + 0.1 r_{jt})\, \alpha_j^{rev} \mathcal{W}\left(h_{jt}, x_{jt}, h_{-jt}, x_{-jt}\right)}_{\text{streamer's revenue}} - \underbrace{\gamma_{jt}(c_1 h_{jt} + c_2 h_{jt}^2)}_{\text{streaming cost}} \tag{1}$$

where h_{-jt} and x_{-jt} denote the streaming hours and states of all streamers other than streamer j in period t. The term $(0.6 + 0.1 r_{jt})$ captures the institutional feature that streamers receive 60% of generated revenue, with an additional 10% if they are approved for the rewards program.

Timeline In each period, the sequence of events is as follows.

1. Streamers draw $\mathcal{I}_{jt}$ and an exogenous exit shock ξ_{jt}. If $\xi_{jt} = 1$, they exit the industry.
2. Streamers draw a scrap value ϕ_{jt} and exit if it exceeds their continuation value.
3. Streamers draw a cost shock γ_{jt} and choose their streaming hours by solving the following Bellman equation (2), conditional on continuation. Because I focus on a steady-state equilibrium, I drop the t subscript and use $'$ to denote the next period:

$$V\left(h_j \mid x_j, h_{-j}, x_{-j}\right) = \max_{h_j} \left[u + \beta \sum_{x'} \mathbb{P}\left(x' \mid x, h\right) \tilde{V}\left(x'\right) \right] \tag{2}$$

[22] Some streamers may also derive non-financial returns, such as attention, from streaming. In the model, this can be captured by lower effective streaming costs.

where x and h denote the states and actions of all streamers, and $\tilde{V}(x') = \mathbb{E}_{\xi,\phi,\gamma,\mathcal{I}}[V(x')]$ is the ex-ante value function.

4. The states $\log(bookmark)_{jt}$ and $score_{jt}$ transition according to their respective processes.
5. The state r_{jt} transitions: some streamers receive the Best Broadcaster status, moving from $r_{jt} = 0$ to $r_{jt} = 1$ permanently.
6. New entrants enter the platform, and the industry transitions to a new state.[23]

3.1.2 Simplifying Assumption

Without simplification, each streamer must take into account the states and actions of all other streamers. Approximately 6,000 streamers appear in the data each month. Thus, even if there are only two states, a streamer would need to evaluate more than 2^{6000} possible configurations to construct a complete contingency plan. To address this computational challenge, I impose the following assumption.

Assumption. Category-level log(bookmark)-weighted sum of streaming hours, ψ_{ct}, fully captures other streamers' influence on streamer j's watch time. Moreover, $\mathcal{W}$ is weakly decreasing in ψ_{ct}.

In mathematical terms, this assumption implies

$$\mathcal{W}(h_{jt}, x_{jt}, h_{-jt}, x_{-jt}) = \mathcal{W}\left(h_{jt}, x_{jt}, \psi_{c(j)t}\right), \tag{3}$$

where h_{jt} is streamer j's streaming hours in period t, $c(j)$ is j's broadcasting category, and ψ_{ct} is defined by

$$\psi_{c(j)t} := \sum_{k \in c(j)} log(\text{bookmark}_{kt}) \cdot h_{kt}.$$

This assumption primarily imposes two restrictions on my model: (1) there is no cross-category interaction, i.e., Game streamers only compete with other Game streamers; and (2) within-category competition is captured by a single aggregate variable, $\psi_{c(j)t}$, while more granular, individual-level information is ignored.

[23] I assume that entry is exogenous and use the observed entry process, denoted by v^e, as described in Section 5.1. Solving streamers' dynamic programs determines transition probabilities across states, which define an industry-level transition matrix M_t. The industry state s_t then evolves according to $s_{t+1} = s_t M_t + v^e$.

These restrictions are supported by descriptive evidence from the empirical setting. First, streamers state that an important determinant of their viewership is whether other popular streamers are broadcasting similar content. Second, XSplit, a commonly used streaming aid tool, advises that it may be a good idea to avoid head-to-head competition with streamers broadcasting similar content.[24] Third, the market is highly fragmented in the data, such that streamers rarely track other streamers' actions at the individual level: even the top 1% of streamer-month observations account for only 2.3% of total monthly watch time.

Finally, it is possible to consider alternative aggregation variables, such as total streaming hours within a category. However, this specification provides a tractable summary of competitive pressure that captures two salient components: streamers' popularity (via bookmarks) and activity (via streaming hours).

3.1.3 Equilibrium

This paper focuses on a symmetric steady-state equilibrium of streamers' dynamic game. Streamers believe that $\{\psi_c\}$, the category-level weighted sum of streaming hours, is fixed at a constant value over time. Under this belief, streamer j's policy $\mu(x_j)$—exit cutoff and streaming hours by state—becomes a function of their own state only.

The equilibrium is defined as the point at which streamers' common belief and the market-level realization of $\{\psi_c\}$ coincide. Formally, the equilibrium is defined as follows:

Equilibrium Definition. *The equilibrium in this paper consists of a policy function $\mu^*(x_j) = \{\rho^*(x_j), h^*(x_j)\}$ and a category-level weighted sum of streaming hours $\{\psi_c^*\}$ that satisfy the following conditions.*

1. *(Optimal Policies) Under the belief that $\psi_{ct} = \psi_c^*$ in the present and future, the policy μ^* satisfies the following:*
   - *(a) h^* solves (2), the streamer's Bellman equation conditional on continuation.*
   - *(b) ρ^* solves the streamer's optimal exit problem, in the sense that $\rho^*(x) = \mathbb{E}_\gamma[V(x)]$. In other words, streamers endogenously exit if and only if the*

[24] https://www.xsplit.com/blog/best-time-to-stream , *"If you're a Tekken (a video game) streamer, maybe it might be good to pick a time that the most popular Tekken streamers aren't on."*, retrieved on June 30, 2023.

scrap value exceeds the expected continuation value.[25]

2. *(Consistency) If all players follow the policy μ^*, $\{\psi_c^*\}$ is consistent with the corresponding expected steady-state action implied by* (6) *and μ^*.*[26]

Methodologically, this equilibrium concept is similar to the Oblivious Equilibrium (OE) proposed by Weintraub et al. (2008, 2010), in that a streamer believes that other streamers' states are permanently fixed and unaffected by their own actions.

As discussed earlier, the streaming platform's environment during the data period supports the assumptions required for applying OE. Market share, measured by watch time, does not exhibit significant concentration among top streamers, and market-level outcomes such as the share of Best Broadcasters and total generated revenue remain roughly constant over time.

Lastly, mapping streamers' optimal policies to long-run, steady-state market-level outcomes requires defining the entry process and its connection to the long-run industry state. In short, I assume that the entry process is fixed at its empirically observed level. Details on computing the long-run industry state are presented in Appendix B.1.

While changes to the streamer reward program may theoretically affect the entry process, I rule out this possibility because new entrants are typically far from obtaining the Best Broadcaster reward, and the growth dynamics of new streamers are highly unpredictable. Moreover, the reward program design has not changed in practice, so there is limited variation to credibly identify the impact of program design on entry.

3.2 Model Discussion

Before proceeding to estimation, the following subsections clarify several features and limitations of the model.

[25] This endogenous exit is orthogonal to exogenous exits caused by the ξ_{it} draw. The former is a streamer's decision based on the expected discounted sum of profits and the scrap value draw. The latter represents a "forced" exit unrelated to profits, such as mandatory military service.

[26] The realization of this weighted sum of streaming hours is stochastic due to i.i.d. shocks drawn by streamers within each period. Streamers consider the expected value of this realization at the beginning of each period.

3.2.1 Tournament Feature

In practice, streamers participate in monthly tournaments to obtain the Best Broadcaster reward. In the model, streamers do not directly consider other streamers' score distribution. Streamers behave as if approval probabilities are exogenously given and fixed based on their own score. This simplification is reasonable if streamers believe that their individual actions do not significantly influence the overall score distribution and ranking, which is plausible given the large and fragmented market.

Indeed, during my interviews with streamers who were recently accepted into the rewards program, I found that they only had a rough idea of their acceptance probability, let alone their exact ranking. One streamer told me that he delayed his application for several months because he was not sure whether his score was sufficient for approval. This was despite his score already being above the typical cutoff. Ultimately, he was accepted on his first attempt.

The model still captures an aspect of the tournament in the counterfactual simulations described in Section 5. At the counterfactual stage, I adjust score-based approval probabilities in an outer loop until the resulting equilibrium aligns with the intended program design (e.g., the number of approved streamers doubles).

This approach allows me to incorporate the dynamic influence of other streamers to some extent. For example, if other streamers respond quickly to an increase in approval likelihood, the number of streamers newly approved at equilibrium will also increase rapidly, causing the increase in approval probability to stop earlier in the outer loop. From a focal streamer's perspective, this may appear as if other streamers are working harder, thereby lowering the streamer's chance of approval in the tournament.[27]

3.2.2 Market Expansion, Stealing, and Network Effects

This model allows for both market expansion and market stealing, but assumes that the latter quantitatively dominates. This restriction is reflected in the assumption that a streamer's watch time decreases when other streamers in the same category stream more (i.e., when the weighted sum of streaming hours ψ_c increases). One can interpret this as ruling out positive network effects both within streamers and

[27]Strictly speaking, however, this paper, like many other structural papers, implicitly assumes a representative agent whose choices depend on their states. This assumption makes it difficult to explicitly separate focal and competing streamers within a single tournament from a modeling perspective.

between streamers and viewers.

This restriction can be justified and is useful for two reasons. First, the streaming market in South Korea is sufficiently mature, exhibiting no significant growth over time during the data period (2019–2020). Thus, rather than expanding the market, competition among streamers is more likely to operate as capturing a larger share of a fixed pie. Second, while imperfect, the restriction alleviates concerns about multiple equilibria.[28]

3.2.3 Focus on Platform Revenue Instead of Profit

While a platform's ultimate objective is to maximize profit, this paper focuses on platform revenue. Specifically, one may want to account for cost changes associated with the reward program design, such as increases in server maintenance costs.

However, such cost changes are likely to be small relative to revenue changes. When the reward program design changes, the main associated cost change is the change in dedicated internet circuit expenses due to changes in watch time. AfreecaTV's 2019 Q4 earnings fact sheet shows that internet circuit expenses amount to about 7.7 percent of revenue generated from commissions on viewer tipping.[29]

4 Estimation

4.1 Overview

First, I specify and calibrate shock distributions and transition dynamics. Second, I approximate watch time as a function of a streamer's states, streaming hours, and category-level streaming hours. Third, I estimate five model parameters: streaming costs c_1, c_2, scrap value distribution parameters K_0, K_1, and the cost shock dispersion parameter σ_γ.[30]

The five structural parameters are estimated using pseudo maximum likelihood. I construct a pseudo log-likelihood based on the probabilities of actions, such as exiting

[28]For example, if a streamer's watch time increases when other streamers stream more, one can easily obtain multiple equilibria: either no one streams, or everyone streams extensively.

[29]The fact sheet is available from https://corp.afreecatv.com/ir.php?page=earning. Specifically, circuit expenses were about 2.3 billion KRW, while revenue from tipping was about 30.2 billion KRW.

[30]These parameters are estimated separately for each category-α^{rev} (profitability) subgroup to allow for flexibility. For example, streamers with higher revenue per watch time α^{rev} may have better communication skills and thus enjoy a higher outside option on average.

the platform and choices of streaming hours.[31] To compute the log-likelihood for each parameter guess, I solve the streamer's dynamic problem using policy iteration under a full-solution method. Finally, I obtain the parameter values that maximize the log-likelihood using a simplex method.

Discretization The platform provides an official coding system for granular broadcasting content. I collapse these codes into three broad categories: video game, social, and others. The first category includes streamers who play mobile or PC video games such as League of Legends. The second category consists of streamers who engage in interactive conversations with viewers, often incorporating dancing and singing. The third category includes streamers with diverse content, ranging from boat fishing and woodworking to stock price prediction.

Next, the continuous state variables—per watch time revenue α^{rev}, log(bookmark), and score—are discretized. Specifically, streamers are grouped within each broadcasting category into three levels (low, medium, high) based on their individual α_j^{rev}, and scores are discretized more finely around the average approval cutoff. Details of this discretization are provided in Appendix B.2.

Furthermore, I discretize monthly streaming hours into $\{0, 50, \cdots, 300\}$ hours. Streamers choose one of these levels depending on their state.

4.2 Parametrization & Calibration

4.2.1 Shocks

I parameterize the distributions of independently and identically distributed shocks as follows.

Exits First, the scrap value $\phi_{jt} \overset{\text{iid}}{\sim} \exp(K_0 + K_1 \cdot bookmark_{jt})$. Second, a multiplicative cost shock $\gamma_{jt} \overset{\text{iid}}{\sim} lognormal(0, \sigma_c^2)$.

Next, the parameters for binary shocks are determined directly from the data. For the exogenous exit shock ξ_{jt}, $\mathbb{P}(\xi_{jt} = 1)$, i.e., the probability of exogenous exit, is 0.0128. While the observed exit probability decreases with popularity (measured by log(bookmark)), it becomes roughly constant at 0.0128 for values in the range $log(\text{bookmark}) \geq 12$.

[31] This approach is pseudo because I fix a theoretically endogenous aggregate variable (the weighted sum of streaming hours at the category level) at its observed average to reduce computational burden. The justification is that streamers have rational expectations about its realization.

This possibility of exogenous exit is useful for two reasons. First, it captures exits not related to revenue (e.g., mandatory military service). Second, it prevents situations in which streamers in certain states almost never exit, causing the number of streamers in these states to become non-stationary.[32]

Application and Approval Probabilities Streamers draw $\mathcal{I}_{jt} \in \{0, 1\}$ and apply to the reward program if $\mathcal{I}_{jt} = 1$. In each period, $\mathbb{P}(application) = \mathbb{P}(\mathcal{I}_{jt} = 1) = 0.067$.[33] This value corresponds to the observed probability that streamers with $score \geq 66$ receive the reward. Because these streamers have scores above the typical cutoff, their probability of receiving the reward approximates $\mathbb{P}(application)$.

Conditional on application, I use the empirically observed approval probabilities by score bin (shown in Figure A.8) in the estimation stage. In the estimation stage, I treat these approval probabilities as exogenous and fixed; in the counterfactual analysis, they are endogenized to align with a given counterfactual reward program design.

4.2.2 Time Discount Factor

I set the discount factor to $\beta = 0.99$. Although the time unit is short (one month), setting β too close to 1 may create computational difficulties, such as slow convergence when solving the dynamic problem. In addition, streamers' popularity is volatile, which may further reduce their effective discount factor.

4.2.3 Transition Dynamics

To describe the evolution of the remaining two time-varying states, $\log(\text{bookmark})_{jt}$ and $score_{jt}$, I use a variant of a linear probability model to capture their dynamics. As streamers broadcast more, they have a higher probability of starting the next period with both a greater number of bookmarks and a higher score. To approximate the dynamics of either moving up or staying, I employ a linear probability model with

[32] In theory, because the scrap value distribution has support on $[0, \infty)$, the endogenous exit probability is strictly positive for all states. However, in practice, I find that without exogenous exits, some states have endogenous exit probabilities as low as 10^{-27}, so streamers in those states almost never exit.

[33] While this number may appear low, it does not imply that streamers do not value rewards. The probability is small because many streamers interested in rewards have already received them and are thus not part of the pool of new recipients. In addition, as discussed in Appendix A.5, 80% of eligible streamers eventually receive the reward.

a single independent variable (streaming hours). Specifically, I assume the following linear probability model for upward transitions:

$$\mathbb{P}\left(\log(\text{bookmark})_{jt} \text{ or } score_{jt} \text{ increases}\right) = \beta_0 + \beta_1 \cdot hour_{jt} + \varepsilon_{jt} \tag{4}$$

where $hour_{jt}$ denotes the streaming hours of streamer j in month t. Conditional on remaining active, there is a fixed probability that the streamer moves down by one interval.[34] This fixed probability is calibrated using the observed probability of moving down, conditional on remaining active. To allow for heterogeneity across state ranges, transition parameters are permitted to vary across interval subgroups. The results are summarized in Table 3.

This specification is intended to ensure that the marginal dynamic gain—arising from a higher probability of transitioning to a better state in the next period—is concave in streaming hours. In contrast, alternatives such as a logit model tend to generate convex marginal gains, which often lead streamers to choose corner solutions.

4.3 Approximation of Watch Time $\mathcal{W}$

I approximate a streamer's watch time $\mathcal{W}(\cdot)$ using a tree-based machine learning model with shape restrictions. Consistent with the simplifying assumption, watch time is a function of a streamer's state, specifically $\log(\text{bookmark})_{jt}$, r_{jt}, broadcasting category, α_j^{rev}, streaming hours, and the weighted sum of category-level streaming hours ψ_{ct}.

More specifically, I use XGBoost (Extreme Gradient Boosting trees) and approximate its predictions with a function that is concave with respect to streaming hours. Details of the watch time approximation, including a comparison of different machine learning models and model fit, are provided in Appendix B.4.

Compared to logit-based models, this approach offers two advantages. First, it delivers superior out-of-sample predictive accuracy, even after imposing shape restrictions. On average, relative to a nested logit model, I observe an improvement of approximately 24% in out-of-sample root mean squared error (Appendix YY).

Second, and more importantly, it allows me to impose concavity of watch time, i.e., diminishing marginal returns to additional streaming hours. Unlike price, a commonly studied choice variable, streaming hours affect market share in a way that

[34] It is possible to allow the probability of moving down to decrease with streaming hours. However, because these probabilities are small, using a linear probability model often yields negative predicted probabilities. As a result, I impose a constant probability.

Table 3. $log(bookmark)_{jt}, score_{jt}$ transition dynamics estimates.

score range	$y = \mathbb{1}\{score\ up\}$		$\hat{\mathbb{P}}(down \mid stay)$
	$\hat{\beta}_0$	$\hat{\beta}_{100hours}$	
$0 - 60$	.0396	.0375	.0541
	(.0016)	(.0023)	
$60 - 66$	.1164	.1277	.1641
	(.0091)	(.0081)	
≥ 66	-	-	.0236

$log(bookmark)$ range	$y = \mathbb{1}\{bookmark\ up\}$		$\hat{\mathbb{P}}(down \mid stay)$
	$\hat{\beta}_0$	$\hat{\beta}_{100hours}$	
$0.0 - 5.0$	.0582	.1346	.0096
	(.0026)	(.0048)	
$5.0 - 8.5$	.0478	.0378	.0081
	(.0024)	(.0024)	
$8.5 - 11.0$	.0195	.0144	.0049
	(.0024)	(.0023)	
≥ 11	.0036	.0039	.0032
	(.0029)	(.0026)	

Notes: Streamers have the opportunity to either move up or remain in their current interval based on their streaming time. Conditioning on staying, there exists a probability of moving down. For readability, the unit of monthly streaming hours is set to 100. The numbers in parentheses are robust standard errors.

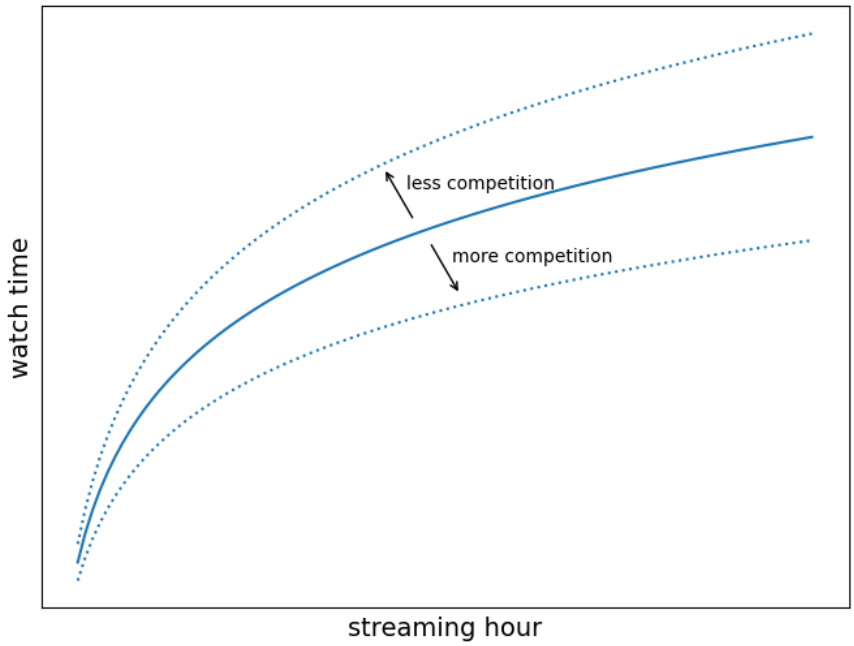


Figure 3. Visualization of the watch time function $\mathcal{W}$ for a given state.

Notes: Conditional on a state x, watch time is increasing and concave in streaming hours. Competition, measured by the category-level weighted sum of streaming hours $\psi_{c(j)t}$, shifts $\mathcal{W}$.

resembles increasing effort. In logit models, streamer j's market share (watch time) is proportional to $s_j = \frac{\exp(h_j \cdot \beta_h)}{\exp(h_j \cdot \beta_h) + C}$, where $\beta_h > 0$ and h_j denotes streaming hours. One can verify that $\frac{\partial s_j}{\partial h_j} > 0$ and increasing in h_j.

This convexity of watch time with respect to streaming hours creates a severe corner-solution problem when solving the model. If a streamer finds it profitable to increase streaming hours from 0 to 50, they will find it even more profitable to increase from 50 to 100, since convexity implies increasing marginal gains. As a result, streamers tend to choose either zero or the maximum possible streaming hours, and the model cannot generate the full range of observed streaming hour choices.[35]

The fitted watch time function $\mathcal{W}$ satisfies the intended shape restrictions: it is increasing and concave in streaming hours, and decreases with competition at the category level. Figure 3 and additional details are provided in Appendix B.4.

I exploit across-month variation in ψ_{ct} to capture how an increase in ψ_{ct} reduces

[35]To address this issue, I also experimented with an alternative "hockey stick" (rather than quadratic) cost function. However, it becomes difficult to specify how the cost shock γ_{jt} perturbs costs. In principle, the shock should affect both the threshold and slope of the hockey stick, but then the shock draw is no longer monotonic in the action (hour choice). Without such monotonicity, one must solve a large-scale mixed-integer programming (MIP) problem at each policy iteration step, which makes policy iteration computationally infeasible.

a streamer's watch time when estimating $\mathcal{W}$. However, in the estimation stage, I assume that all observations come from a single steady-state equilibrium of the dynamic game, where ψ_{ct} is fixed at its observed monthly average. Therefore, this approximation does not materially affect the estimation.

4.4 Pseudo Maximum Likelihood Estimation

Given the previously approximated model components, I employ a full-solution approach to estimate the structural parameters. For each parameter guess, I solve the streamer's dynamic programming problem using policy iteration and compute the log-likelihood based on the predicted conditional choice probabilities of exit and streaming hours for streamer j in month t.[36]

Let $\hat{\mathbb{P}}^{exit}_{jt}$ denote the predicted probability that streamer j exits in period t, and $\hat{\mathbb{P}}^{h}_{jt}(h_{jt} \mid x_{jt})$ denote the predicted probability of choosing the observed streaming hours h_{jt} conditional on not exiting.

$$\begin{aligned}\log L_{jt} = & \underbrace{\mathbf{1}\{exit\}_{jt} \cdot \log \hat{\mathbb{P}}^{exit}_{jt}}_{\text{exit observed}} \\ & + \underbrace{(1 - \mathbf{1}\{exit\}_{jt})}_{\text{no exit observed}} \cdot \left[\underbrace{\log(1 - \hat{\mathbb{P}}^{exit}_{jt})}_{\text{stay probability}} + \underbrace{\log \hat{\mathbb{P}}^{h}_{jt}(h_{jt} \mid x_{jt})}_{\text{hours choice}} \right]\end{aligned} \tag{5}$$

The identification of structural parameters comes from a number of variations. First, an individual streamer shows different actions around being approved to the rewards program and getting commission discount. This local variation helps identifying c_1, c_2, because the same individual faces time varying marginal gains from streaming. Second, streamers having different states exhibit different hour and exit decisions, so this across individual variation help identify all parameters $c_1, c_2, K_0, K_1, \sigma_c$.[37]

4.5 Estimates

Table 4 presents the estimates of $\{c_1, c_2, K_0, K_1, \sigma_c\}$. There is substantial heterogeneity both across and within broadcasting categories. First, within a category,

[36] Strictly speaking, I do not fully solve the dynamic game among streamers, as I fix the equilibrium object—the category-level streaming hours—at its observed level. This restriction is relaxed when I conduct counterfactual simulations.

[37] Streamer heterogeneity that is not captured by my model may disrupt this argument. However, I explicitly control observed state variables, bookmark and score, and in addition control per watch time revenue α^{rev}_j which could capture some unobserved characteristics like communication skill to mitigate this concern.

streamers with high α_j^{rev} values appear to have higher streaming costs, which may reflect greater effort invested in preparing broadcasting content to attract viewers. In addition, their superior ability may imply higher outside options.

Second, heterogeneity across categories may be driven by differences in the psychological costs of producing broadcasting content. For example, while playing (video) games may be enjoyable, producing content based on social interactions may be more nerve-wracking. This is because such streamers are often engaged in greater emotional labor and care about maintaining a "good" image. When a Social streamer's relationship status is publicly revealed, some viewers often leave.

While streamers with higher α_j^{rev} values may have stronger incentives due to the rewards program's 10% commission discount, their higher streaming costs may partially offset this effect. In the counterfactual simulations, the cost parameters determine which types of streamers are more responsive to changes in the rewards program design, potentially crowding out other types. These heterogeneous responses shape the composition effects discussed in the introduction.

4.6 Model Fit

This section evaluates model fit through a factual simulation and discusses the sources of discrepancies between model predictions and observed data. I first conduct a factual simulation using the full equilibrium search algorithm described in Section 5.1. That is, the environment—including approval probabilities by score—remains unchanged, but ψ_{ct} is endogenized. During the parameter estimation stage, ψ_{ct} was fixed at its observed average level to reduce computational burden.

This factual simulation shows how well the framework approximates the data. To evaluate model fit, I compare predicted and observed outcomes in terms of streaming hours and generated tipping revenue. Figure 4 shows that the model predictions closely match the observed values, suggesting that the model provides a reasonable approximation of streamers' behavior.

Further assessments of model fit are provided in Appendices C.1, C.2, and C.3, in addition to Figure 4. Specifically, I compare predicted and observed industry-level variables such as the distribution of streamers' bookmarks, evaluate whether the reward cutoff moves similarly to the model's prediction following AfreecaTV's design change implemented outside the sample period, and discuss factors that may contribute to the remaining gaps.

In the next section, I conduct counterfactual simulations to evaluate alternative

Table 4. Pseudo Maximum Likelihood Estimates.

category	α^{rev}	c_1	c_2	K_0	K_1	σ_c
Game	low	0.290	0.426	85.313	19.291	1.537
		(0.003)	(0.004)	(0.925)	(0.295)	(0.018)
	med	0.011	0.424	51.011	29.371	1.269
		(0.000)	(0.001)	(0.366)	(0.071)	(0.006)
	high	8.159	1.186	245.709	108.628	1.206
		(0.059)	(0.008)	(2.966)	(0.486)	(0.015)
Social	low	0.117	1.919	660.712	31.758	1.726
		(0.002)	(0.044)	(14.306)	(0.436)	(0.031)
	med	0.018	5.841	643.550	223.352	1.327
		(0.000)	(0.038)	(4.029)	(1.835)	(0.006)
	high	1.178	10.282	801.125	565.757	1.039
		(0.003)	(0.011)	(16.358)	(1.667)	(0.001)
Other	low	0.013	0.015	11.328	0.312	1.695
		(0.000)	(0.000)	(0.159)	(0.003)	(0.018)
	med	0.039	0.304	60.946	12.888	1.536
		(0.001)	(0.003)	(0.066)	(0.151)	(0.004)
	high	0.014	1.929	240.851	110.615	1.138
		(0.000)	(0.001)	(0.763)	(0.074)	(0.001)

Notes: The numbers in parentheses represent bootstrap standard errors ($B = 20$). The unit of model parameters is approximately 1 USD, except for K_0 (1,000 USD for readibility) and σ_c (unitless).

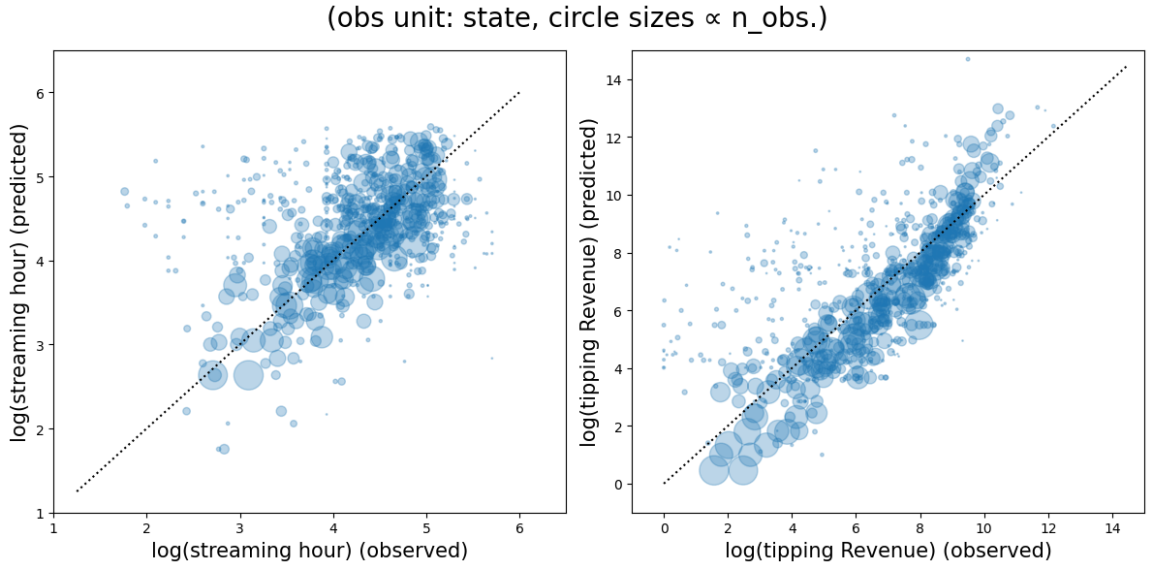


Figure 4. Predicted versus observed streaming hours and tipping revenue.

Notes: The unit of observation is a streamer's state, i.e., a combination of category, profitability, bookmarks, score, and Best Broadcaster status. Each value is averaged at the state level, and the size of each circle is proportional to the number of observations.

rewards program design. When doing so, the counterfactual outcomes are always compared with factual simulation outcomes.

5 Counterfactual Simulations

I conduct counterfactual simulations to assess changes in the design of the rewards program. First, I vary the number of monthly approval slots by doubling and halving it. Second, I change the existing 10% commission discount for approved streamers to 5% and 15%. Lastly, I reallocate approval slots at the category level, assigning two additional slots to the Social category and removing two slots from the Game category.

The primary purpose of these counterfactual exercises is to study whether platform revenue can be improved through changes in the rewards program design. The first two simulations examine how changes in the number of approval slots and commission discounts affect platform revenue. The third simulation investigates whether a more granular program design—assigning rewards at the category level rather than the platform level—can improve platform revenue.[38]

[38]Additionally, this third simulation helps identify design changes that could improve platform revenue without negatively affecting streamers overall. While the simulation results support reduc-

Decomposing Reward Design Change Effects Platform revenue is the product of three components: total watch time, average revenue per unit of watch time, and the platform's share of total revenue. Accordingly, the overall effect on platform revenue can be decomposed into changes in these three components. This decomposition provides a more granular understanding of how reward program design affects platform revenue.

To summarize, reducing the number of monthly approval slots or the commission discount appears to be more effective in increasing platform revenue. Providing more rewards motivates streamers—especially more profitable ones—to stream more, increasing total watch time (*watch time effect*) and improving average revenue per unit of watch time (*profitability effect*). However, the reduction in the platform's share of tipping revenue due to higher commission discounts (*platform share effect*) generally outweighs these gains.

5.1 Full Equilibrium Search

When searching for a full equilibrium under different environments—i.e., varying the number of reward program slots and the level of benefits (commission discount %)—a key computational challenge is that it is not straightforward to embed the tournament design into a streamer's dynamic programming problem.

I take an indirect approach. The approval probabilities by score, as described in Section 4.2.1, are adjusted until a desired steady-state outcome is achieved. For example, to double the number of slots, approval probabilities are increased until the number of streamers receiving the reward doubles at the steady state relative to the factual simulation. I first present the full equilibrium search algorithm, then provide further details on its components (e.g., the entry process), and finally describe the results.

I first clarify the notation used in the algorithm described below. N_{slot} represents the target number of slots, $\hat{N}_{slot}$ denotes the predicted number of approved streamers at equilibrium, and N_{slot}^{gap} represents the difference between these two values. $\{\psi_c\}$ denotes the category-level weighted sum of hours.

In the context of the algorithm, the term "marginal" score group refers to the

ing the scope of the program, including rescinding benefits from already approved streamers, such changes may be difficult to implement in practice. Bewley (1998) shows that, although pay cuts may be a reasonable response to changes in the economic environment, they are rarely implemented because managers believe they harm employee morale.

highest (lowest) score group for which the approval probability is less than 1 (greater than 0) when approval probabilities are increased (decreased). It is implicitly assumed that changes in the number of slots primarily affect the approval probability of this highest/lowest score group.

Algorithm Full Equilibrium Search

```
Specify the commission discount percentage.
Specify the number of slots N_slot.
Initialize approval probabilities by score.
N_slot^gap, ψ_c^gap ← 1

while N_slot^gap > ε^N do
    Increase or decrease the approval probability of the "marginal" score group.
    Initialize {ψ_c^old} (weighted sums of hours).

    λ ← 1
    while ψ_c^gap > ε^ψ for any c do
        Given {ψ_c^old}, solve a streamer's DP and obtain optimal policies.
        Compute the corresponding industry state and ψ_c^new (cf. equation (6)).
        ψ_c^gap ← ||ψ_c^new − ψ_c^old||_1
        ψ_c^old ← ψ_c^old + (1/√λ) · (ψ_c^new − ψ_c^old); λ ← λ + 1
    end while

    From the equilibrium industry state, compute N̂_slot.
    N_slot^gap ← |N_slot − N̂_slot|
end while
```

Notes: "marginal" score group refers to the highest (lowest) score group for which the approval probability is less than 1 (greater than 0) when approval probabilities are increased (decreased).

Additionally, I provide remarks on two details of this equilibrium search algorithm.

Entry Process I use the empirical entry process observed in the data. The data begin in October 2019, but one streamer-tracking website started monitoring streamers in January 2019. Therefore, I can identify streamers who newly appear each month. For example, I observe that, on average, 1.29 streamers with the lowest level of bookmarks and score, non-Best Broadcaster status, in the Social category, and with low profitability enter the market each month.

This entry process is assumed to remain unchanged throughout the counterfactual simulations. While the entry process itself may respond to reward program designs,

it is likely to have a limited impact on the simulation outcomes, given that many entrants are not popular streamers and have little chance of receiving the reward.

Endogenizing ψ_{ct} During the estimation step, I assume that the category-level weighted sums of hours, denoted by ψ_{ct}, which capture streamer interactions, are fixed at their observed averages to reduce computational burden.

This restriction is relaxed during the counterfactual stage. When a specific reward program design is given—i.e., the number of approval slots and the commission discount percentage—the algorithm proceeds as follows. In the outer loop, I adjust approval probabilities by score. In the inner loop, I search for the corresponding equilibrium and industry state. I iterate on the outer loop until the number of streamers newly approved at equilibrium coincides with the given number of slots. The commission discount percentage remains fixed throughout the algorithm, but it can be set to, for example, 5 or 15 percent instead of 10 percent (status quo) to evaluate alternative program designs.

Note that, to find an equilibrium in the inner loop, it suffices to find the category-level weighted sums of streaming hours at equilibrium, denoted by ψ_c^*. Once these weighted sums are given, the only remaining step is to solve a streamer's single-agent dynamic programming problem.

Before concluding this subsection, I clarify two limitations of the full equilibrium search algorithm and the implementation of counterfactuals. First, the algorithm indirectly assesses tournament design changes only through changes in approval probabilities that are based on score. Second, the probability of applying for the rewards program ($\mathbb{P}(\mathcal{I}_{jt} = 1)$) is assumed to remain unchanged. While approval probabilities can increase in the outer loop, this assumption serves as a cap on the ex-ante approval probability.

5.2 Counterfactual 1: The Number of Monthly Approvals

I first examine a counterfactual reward program design that involves doubling and halving the number of monthly approval slots. Overall, the simulation results show that the former decreases platform revenue by 2.53%, while the latter increases it by 3.03%. Offering the reward to more streamers increases both total watch time and the share of profitable streamers, but the resulting decline in the platform's share of total revenue is large enough to generate an overall negative effect.

As discussed in the overview of this section, the total effect of this design change

can be decomposed into three components. Computing the total watch time effect is straightforward. I compare total watch time before and after the design change. To compute the profitability effect, I first calculate total revenue based on the changed watch time, holding fixed the share of each category-α^{rev} group at the level in the factual simulation. This value is then compared with the total revenue from the counterfactual simulation.

Similarly, to isolate the platform share effect, platform revenue is computed based on the changed total revenue, holding fixed the shares of both the platform and streamers. This value is then compared with the platform revenue in the counterfactual simulation. Table 6 shows the changes in composition and in the platform/streamer shares in each counterfactual simulation.

5.3 Counterfactual 2: Changing Rewards Program Benefit

Second, I examine a counterfactual platform policy that offers either a 5% or a 15% commission discount (instead of the status quo of 10%) when a streamer is approved. Using the same decomposition procedure described above, I compute the overall effect and present the results accordingly.

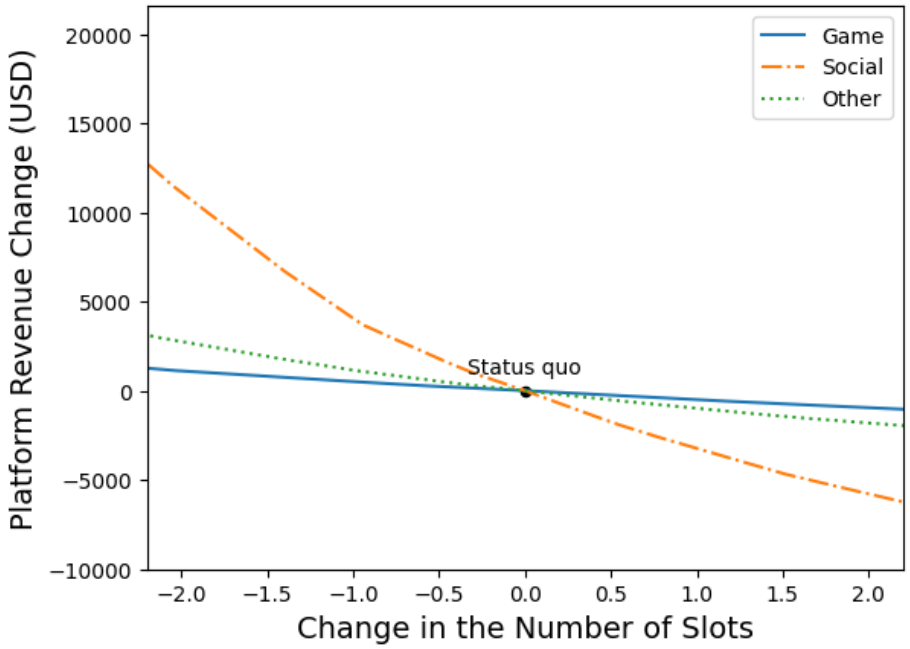
Platform revenue increases by 6.08% when the commission benefit is reduced. The watch time and composition effects decrease platform revenue in response to the reduced approval benefit, but the platform share effect dominates these opposing effects. These effects operate in the opposite direction when the commission discount increases to 15%, resulting in a 6.24% decrease in platform revenue.

5.4 Counterfactual 3: Reallocation of Slots at the Category Level

Lastly, I investigate whether platform revenue can be improved by adopting a more granular program design, even without reducing the overall scale of the program. This perspective focuses on the fact that the current program determines the number of slots at the platform level, while the marginal effect of adding one slot can differ across broadcasting categories.

It is straightforward to apply the full equilibrium search algorithm at the category level because there are no cross-category interactions once approval probabilities by score are given. I run the algorithm at the category level and examine how platform

Figure 5. Platform revenue changes when allocating more or fewer slots to specific categories. The black dot in the middle represents the status quo.



revenue changes when allocating one more or fewer slots to each category relative to the status quo.

The marginal change in platform revenue with respect to one additional reward program slot differs substantially across categories (Figure 5). While platform revenue decreases in all categories when offering more slots, the slopes of these revenue changes differ across categories. Therefore, platform revenue can be improved by decreasing the number of slots in the Social category (where the loss from providing rewards is the greatest) and increasing slots in other categories.

I conduct this simulation by reducing two slots in the Social category and adding two slots to the Game category. The number of slots in the Other category remains unchanged. In the status quo, there are 12.19, 4.58, and 4.59 streamers newly approved each period in the Game, Social, and Other categories, respectively. After the reallocation, these numbers change to 14.19, 2.58, and 4.59, respectively.

This reallocation increases platform revenue by 1.19%. Note that streamers in the Social category tend to have higher revenue per unit of watch time. Thus, providing them with a commission discount generates larger program costs through reduced platform commissions, which drives these results. Again, the results are summarized in Table 5 and Table 6.

Table 5. Decomposition of the effect of an alternative design on platform revenue.

(Units: %, relative to the current program design)

Counterfactual	Watch time effect	Profitability effect	Platform share effect	Overall effect
1. The number of streamers getting the reward each period				
doubled	0.22	0.94	-3.66	-2.53
halved	-0.75	-0.08	3.90	3.03
2. Benefit change (current commission discount: 10%)				
increase to 15%	0.07	0.34	-6.63	-6.24
reduce to 5%	-0.10	-0.32	6.52	6.08
3. More granular (category level) slot reallocation				
Social to Game	0.02	-0.45	1.62	1.19

Notes: Watch time, profitability, and 'platform share effects refer to the platform's revenue gain resulting from changes in total watch time, average per watch time tipping revenue, and the platform's share among tipping revenue, respectively..

Table 6. Detailed streamer decomposition (above) and platform's share change (below).

(Panel A: Composition Effect Details)

category	α^{rev}	Watch Time Share Compositions (%)					
		Status Quo	Doubling Slots	Halving Slots	5%p more commission cut	5%p less commission cut	Slot Reallocation
Game	low (0.017)	23.27	23.07	23.18	23.35	23.16	23.23
	med (0.150)	20.83	21.20	20.48	20.95	20.72	20.92
	high (0.698)	16.87	17.01	16.84	16.95	16.79	16.91
Social	low (0.239)	1.46	1.44	1.48	1.45	1.47	1.46
	med (1.591)	3.94	3.98	3.96	3.94	3.95	3.91
	high (5.192)	7.66	7.72	7.67	7.68	7.64	7.60
Other	low (0.012)	11.00	10.45	11.48	10.66	11.34	11.00
	med (0.205)	6.98	7.01	6.99	6.98	6.99	6.98
	high (1.573)	7.99	8.11	7.93	8.04	7.94	7.99

(Panel B: Platform Share Change Details. All numbers are percentage)

Counterfactual	Status Quo	Doubling Slots	Halving slots	5%p more commission cut	5%p less commission cut	Slot Reallocation
Streamer's share	64.59	65.88	63.21	66.94	62.28	64.12
Platform's share	35.41	34.12	36.79	33.06	37.72	35.88
Total	100.00	100.00	100.00	100.00	100.00	100.00

Notes: The numbers in parentheses represent discretized values of α^{rev}. The unit of measurement is approximately 1 USD per watch time.

Lastly, one potential concern is whether streamers can switch categories in response to this slot reallocation. However, such switching behavior is unlikely to occur frequently because the human capital required for each category is difficult to acquire. For example, to be successful in the Social category, a streamer typically needs to be physically attractive, have strong speaking skills, or both. In the Game category, viewers are usually fans of specific video games, so a deep understanding of those games is required. Moreover, switching categories could lead to a substantial loss of existing viewers (who prefer the current content), causing them to leave. Thus, streamers are unlikely to switch categories freely.

6 Conclusion

I develop an empirical dynamic game model to assess the impact of seller rewards programs on platform revenue in the context of livestreaming platforms. The design of rewards programs influences platform revenue through three key channels. First, it affects the working hours of the average streamer (a seller in this industry) and, consequently, the total market size (watch time). Second, it changes the share of watch time allocated to streamers who can extract more or less tipping revenue from a given amount of watch time. This, in turn, affects average profitability, measured as tipping revenue per unit of watch time. Lastly, it affects the platform's share of total tipping revenue, as the platform offers a permanent commission discount as the reward program benefit.

Counterfactual simulations suggest that the platform share effect quantitatively dominates in determining net platform revenue. This finding implies that reducing the rewards program—either by offering the reward to fewer streamers per month or by decreasing the commission discount—would increase net platform revenue. If such unilateral reductions are practically difficult, I additionally show that reallocating program approval slots at a more granular level (i.e., at the broadcasting category level rather than the platform level) may still improve net platform revenue.

One may ask why the platform maintains a seemingly suboptimal reward program design. I offer two possible explanations. First, path dependence. The current program has remained largely unchanged since the platform's launch in 2007. In the early stages, attracting more streamers—even at the cost of lower platform revenue—may have been a sensible strategy to address the "cold start" problem. Once established, such a program may be difficult to scale back. Second, the model does not explicitly

account for competition across platforms. The presence of competing platforms (e.g., Twitch Korea in this context) may induce AfreecaTV to provide additional benefits to retain popular streamers.

Several limitations of this methodology suggest directions for future research. First, the current model assumes that streamers draw i.i.d. shocks when deciding whether to apply to the rewards program, but allowing for unobserved preferences that are correlated over time would be more realistic. Second, the model does not account for competition across platforms or streamer multihoming.[39] Lastly, the model abstracts from an endogenous entry process. Incorporating these features would be a promising direction for future research.

[39] One observation that may support this simplification is that popular content on the focal platform (AfreecaTV) and its rival platform (Twitch) differs substantially, so they could be considered separate markets.

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Online Appendix for
"A Structural Model of Rewards Programs on Digital Platforms: The Case of Livestreaming"

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April 5, 2026

A Additional Details on the Data

A.1 Data Construction Details

The main dataset comes from three sources. From streamer tracking website 1 (poong.today), I collect streamer-day-level revenue and streamer-month-category-level streaming hours. Figure A.6 provides a snapshot of this website. From streamer tracking website 2 (rank.afreehp.kr), I collect streamer-day-level streaming hours, bookmarks, and watch time. Lastly, from the platform's official website, I collect the list of streamers who were accepted into the rewards program each month, as well as the conversion table used to compute streamers' scores.

Data from October 2019 to April 2020 are used because website 2 started recording tipping revenue in October 2019, and the COVID-19 outbreak from spring 2020 had sizable effects on this streaming market. COVID-19 substantially and persistently increased both viewership and the number of streamers, making it difficult to fit the observed patterns within a steady-state framework.

I merge the two datasets from websites 1 and 2 based on the streamer's account name and day. Some observations are dropped because the two websites do not track the exact same set of streamers. However, the merged dataset still covers 90.5 percent of the total revenue recorded on website 1, as both websites tend to track relatively popular streamers more comprehensively.

Next, I add each streamer's score and a dummy variable indicating whether the streamer has already been approved, based on information collected from the platform's official website. Streamers' scores are computed using the platform's official

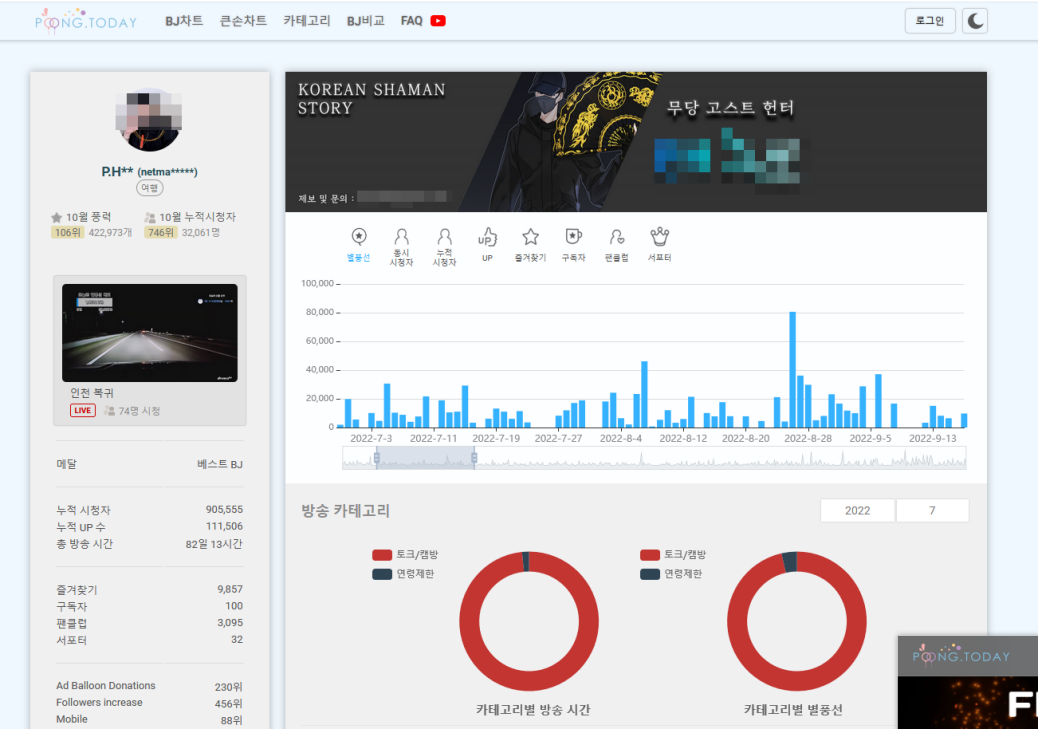


Figure A.6. Snapshot of a tracking website.

Notes: The figure shows a snapshot for one streamer. Blue bars represent daily tipping revenue, and pie charts represent monthly streaming hours by category.

conversion table (see Table A.7 below). The dummy variable is constructed from the monthly approval list.

Next, streamers who always broadcast more than 360 hours per month are dropped. These streamers account for 0.89 percent of observations and are not individuals but organizations such as television news channels. In addition, streamer-month observations are dropped if a streamer exhibits zero streaming hours in all subsequent periods, which is treated as exit. Finally, observations are dropped if streaming hours, score, or bookmark data are missing. These latter two steps drop 13.52 and 14.02 percent of observations, respectively. Finally, I construct an exit dummy variable, which takes the value of 1 if a streamer does not appear in the next month.

A.2 Data Coverage Assessment

To assess the coverage of the collected dataset, I compute platform revenue from the data and compare it with official financial reports posted on the platform's website. The main dataset covers 61.3% of official tipping revenue in the fourth quarter of 2019.[1] The gap arises because streamer tracking websites cannot capture all tipping activity. This limitation arises for two reasons: (1) the servers of tracking websites may fail to record all tipping events when many viewers tip simultaneously, and (2) some streamers are too small to be tracked.

[1]Specifically, for the fourth quarter of 2019, platform revenue (from tipping) computed from the data is 18.5 billion KRW, while the financial report reports 30.2 billion KRW.

A.3 Score conversion table

Table A.7. The score conversion table on AfreecaTV.

score	average viewership (40%)	bookmarks (40%)	total broadcasting hours (20%)
100	(more than) 1000	50,000	10,000
98	750	40,000	9,500
96	500	30,000	9,000
94	450	20,000	8,500
92	400	10,000	8,000
90	350	9,000	7,500
88	300	8,000	7,000
86	250	7,000	6,500
84	200	6,000	6,000
82	160	5,600	5,500
80	130	5,200	5,000
78	100	4,800	4,500
76	90	4,400	4,000
74	80	4,000	3,500
72	70	3,600	3,000
70	60	3,200	2,600
68	50	2,800	2,200
66	40	2,400	1,800
64	30	2,000	1,400
62	20	1,500	1,000
60	10	1,000	500

Notes: The variable average viewership is computed based on a 3-month window. Bookmarks represent the number of viewers who have bookmarked the streamer. The score is a weighted sum of scores from three items. Below the bottom row, the score for each item is zero. The source of this table is https://afevent2.afreecatv.com/app/star_bj/bestbj/order_info.php, retrieved on July 15, 2023.

A.4 Empirical Patterns for Profitability

To understand the factors determining profitability, measured by revenue per unit of watch time, I conduct a set of fixed-effects regressions.

Panel A of Table A.8 shows that individual (streamer) fixed effects explain a substantial portion of the variation. This result is intuitive, since tipping revenue is heavily influenced by personal traits such as communication skills and appearance.

To examine my modeling assumption that profitability is persistent within individuals over time, particularly after receiving Best Broadcaster status, I regress individual streamer–month-level per-watch-time revenue on streamer and month fixed effects, as well as bookmarks, score, and Best Broadcaster status.

The first set of results in Panel B of Table A.8 suggests that per-watch-time revenue may increase after approval. However, the change appears to be driven by a short-run "congratulation effect," meaning that viewers temporarily tip more money to congratulate the approval. In the second set of regressions, I use a subsample that excludes the month before and the month after an approval and find that the changes in per-watch-time revenue across approvals are not statistically significant.

Additionally, I investigate how the watch time composition of streamers with heterogeneous profitability affects total tipping revenue, and whether this composition changes sharply during the data period. To do this, I divide streamers into nine groups. First, I categorize them into three broadcasting categories (Game, Social, and Other). Then, within each category, I rank streamers based on their individual streamer-level per-watch-time revenue and evenly split them into low-, medium-, and high-profitability subgroups.

Figure A.7 shows that watch time composition significantly affects tipping revenue, and there is no sharp change in composition during the data period. Within each broadcasting category, although highly profitable streamers do not have a dominant share of watch time, their tipping revenue is much greater than that of the other two groups.

Table A.8. Analysis of individual streamer-month level per watch time revenue

(Panel A: Explanatory Power of Fixed Effects)

$Y = \log(\text{per-watch-time revenue} + 1)$				
R^2	.852	.021	.057	.853
Adjusted R^2	.816	.021	.057	.817
streamer FE	✓			✓
month FE		✓		✓
additional controls			✓	✓

(Panel B: Approval Effects on Per-Watch-Time Revenue)

$Y = \log(\text{per-watch-time revenue} + 1)$	full sample			subsample		
$\mathbb{1}\{BestBroadcaster\}$ ($\hat{\beta}$)	.041	.051	.048	.020	.031	.025
$\mathbb{1}\{BestBroadcaster\}$ (s.e.)	(.015)	(.015)	(.015)	(.026)	(.026)	(.026)
streamer FE	✓	✓	✓	✓	✓	✓
month FE		✓	✓		✓	✓
additional controls			✓			✓

Notes: The unit of the dependent variable Y is USD per unit of watch time. In Panel B, the subsample excludes observations from one month before to one month after an approval to remove temporary "congratulation" effects. Additional controls include bookmarks, score, and $\mathbb{1}\{BestBroadcaster\}$. To account for skewness, I apply a log transformation and drop the top 2.5% of observations. The results are similar when the log transformation is not applied or when alternative cutoffs (5% or 1%) are used.

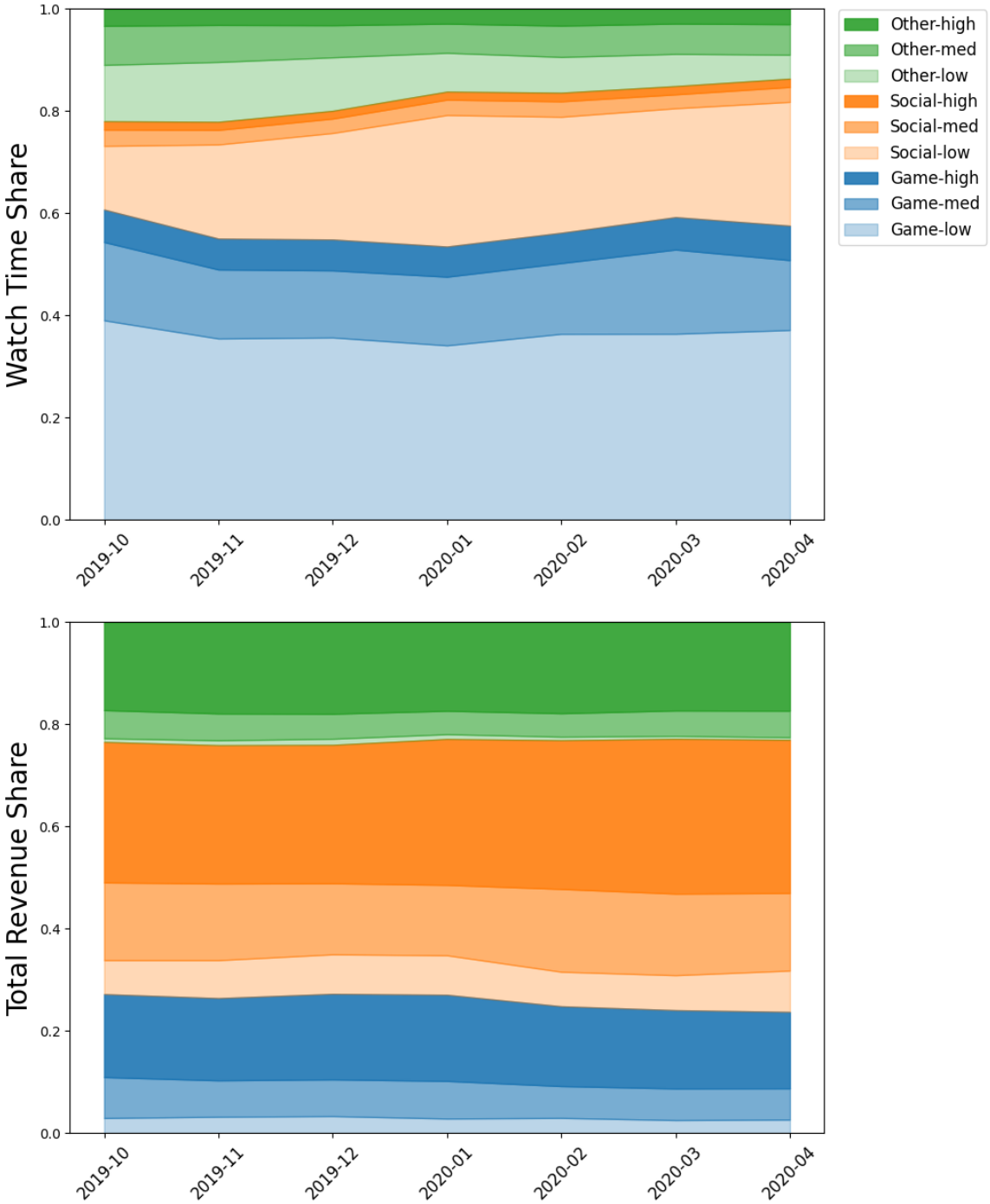


Figure A.7. Watch time and revenue shares by streamer profitability

Notes: Profitability is measured as revenue per unit of watch time. The first label denotes broadcasting categories, while the second label indicates whether streamers have low, medium, or high profitability within each category.

A.5 Additional Patterns in Application Decisions

Long-run application patterns I track streamers who were eligible for the reward program during the data period for three years and find that approximately 80% of them eventually receive the reward. Specifically, 981 streamers were eligible during my data period (Oct 2019–Apr 2020). I track them until Feb 2023. Among those who did not exit, 80% (506 out of 628) eventually received the reward. Therefore, despite frequent delays, the majority of AfreecaTV streamers eventually receive the reward in the long run.

Reasons for delay To investigate the reasons for this delay, I conducted interviews with several streamers on the platform. I contacted streamers who had received the reward using the platform's messaging function. I asked them (1) whether they knew they would be accepted when applying, and (2) what reasons they believe explain why some popular streamers do not apply for the reward program. The interviews suggest the presence of both behavioral and strategic factors. Some streamers were initially uncertain about their chances of approval despite having scores above the typical threshold, but they eventually applied and succeeded on their first attempt. Others cited strategic considerations related to multihoming as a reason for delaying their application. Once participating in the rewards program, streamers are not permitted to livestream simultaneously on other platforms.[2]

Finally, Figure A.8 visualizes the probability of receiving the reward by score group.

[2]Nevertheless, streamers are still allowed to post recorded videos on other platforms or broadcast on other platforms at different times.

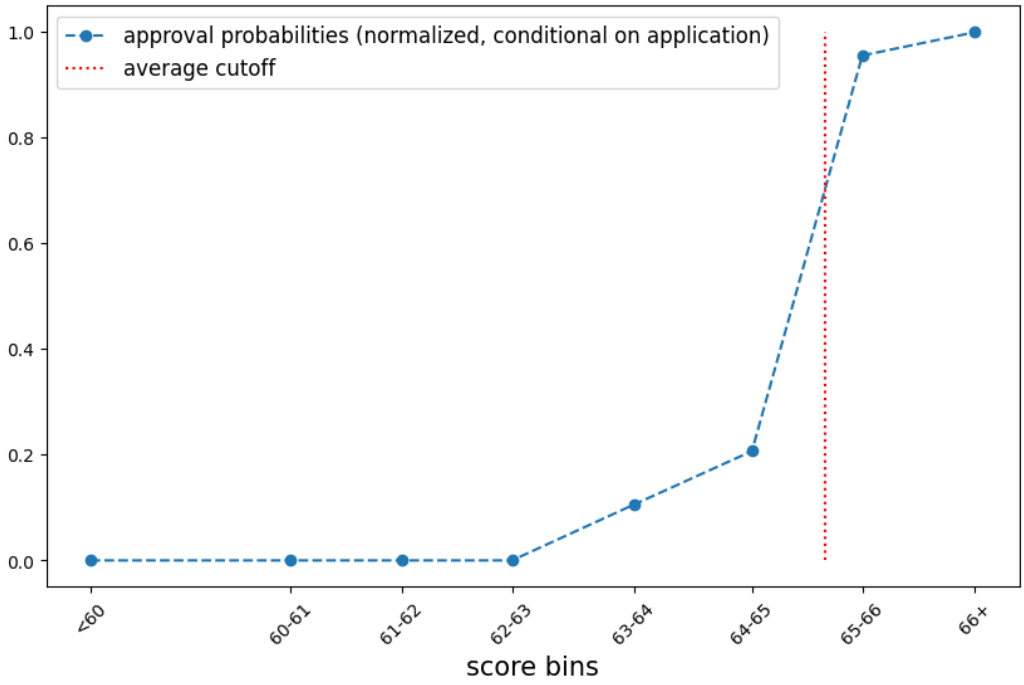


Figure A.8. Probabilities of being approved by the reward program by score range.

Notes: The probabilities were calculated based on observed probabilities of newly receiving the reward. The probability for the *score* ≥ 66 group was set to 1 because they will receive the reward if they apply, while the probabilities for other groups were normalized accordingly.

B Computation Details

B.1 Entry Process and Long-Run Industry States

To compute an equilibrium, one must map streamers' streaming hour choices into the corresponding long-run realization of the weighted sum of streaming hours, ψ_c.

To do so, additional information is required: the entry process of new streamers and the distributions of private shocks. I use the empirically observed entry process, i.e., the number of new streamers entering each state in the data each month, averaged across months during the sample period. Shock distributions (e.g., health issues that perturb streaming costs) are parameterized and calibrated in 4.2, and are assumed to be invariant over time and across reward program designs.

Given this additional information, a policy μ determines an industry-level transition matrix, $M_{\xi,\mathcal{I},\mu}$. Combined with an entry process vector, v^e, the corresponding long-run industry state is given by

$$s_{\xi,\mathcal{I},\mu} = v^e + v^e M_{\xi,\mathcal{I},\mu} + v^e M^2_{\xi,\mathcal{I},\mu} + \cdots = v^e (I - M_{\xi,\mathcal{I},\mu})^{-1} \tag{6}$$

where I denotes the identity matrix. This steady state, together with the policy μ, determines how many streamers are in each state and how many hours they stream. Thus, it is straightforward to compute the corresponding $\psi_{ct} \equiv \mathbb{E}_{\xi,\mathcal{I},\gamma}\left[\sum_{k\in c} \log(\text{bookmark})_{kt} \cdot h_{kt}\right]$, where the expectation is taken with respect to i.i.d. shocks drawn by streamers each period.

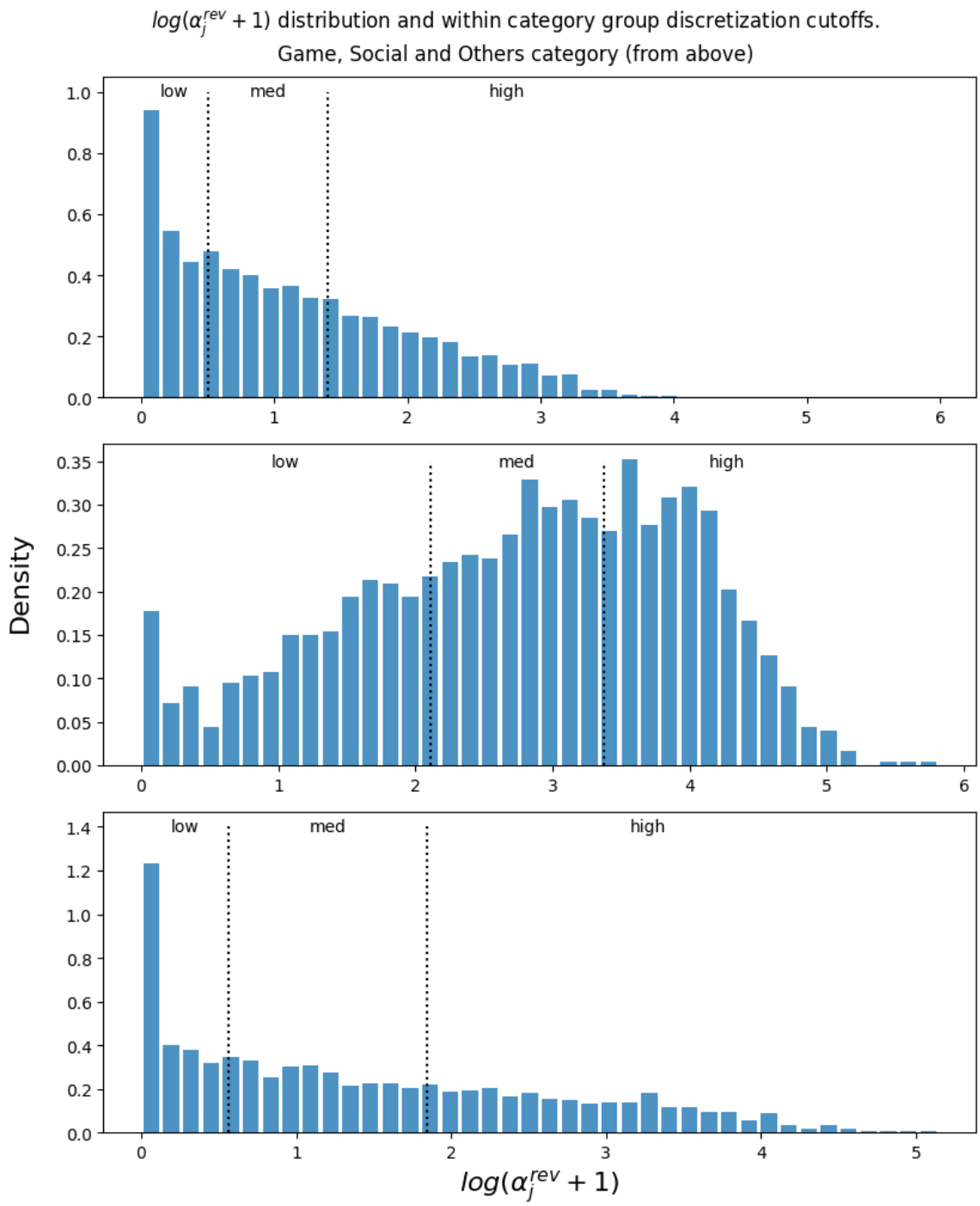
To recap, the equilibrium is defined as the point at which the realized aggregate variable $\{\psi_c^{new}\}$ coincides with the initial belief $\{\psi_c^{old}\}$. A formal definition of the equilibrium concept is provided in 3.1.3.

B.2 Discretization of Continuous Variables

I divide streamers in each broadcasting category into three groups (low, medium, high) based on their individual level α_j^{rev}. Figure B.9 displays the distribution of α_j^{rev} in each broadcasting category and the corresponding cutoffs. For readability, I apply a log transformation.

Next, I discretize the state variables that evolve over time. The binary variable $r_{jt} = \mathbb{1}\{Best\ Broadcaster\}_{jt}$ clearly needs not to be discretized further.I discretize the using the following cutoffs: $[4.0, 4.5, \cdots, 12.5, 13.0]$. Lastly, I discretize the streamer *score* ranging from 0 to 100 using the following cutoffs: $[20, 40, 60, 61, 62, 63, 64, 65, 66]$.

Figure B.9. Per watch time revenue α_j^{rev} distributions within each category



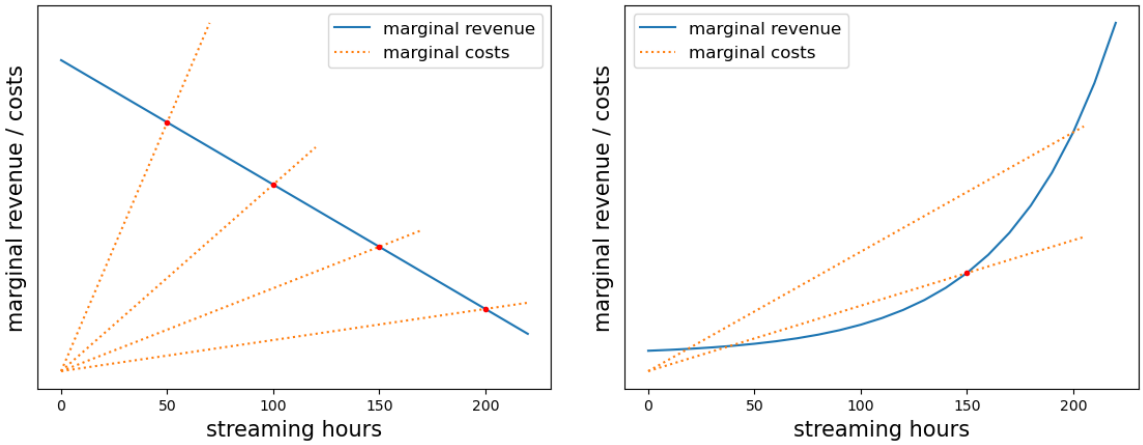


Figure B.10. Concave vs. convex marginal revenue and choice of streaming hours. When marginal revenue is decreasing in streaming hours, multiplicative cost shocks can rationalize interior choices over a wide range of hours (left). In contrast, when marginal revenue is increasing, some choices (e.g., 150 hours) are strictly dominated and are never optimal for any realization of the cost shock (right).

This decision is based on approval cutoffs during the 7-month data period were observed to be around 63.5-65. Therefore, these scores are discretized more granularly around the cutoff value. Below 60, the approval probability is essentially zero, but they were discretized sparsely to capture the rate of streamer growth. Finally, for scores above 66, streamers are all highly likely to get approved if they apply.

B.3 Computational Usefulness of Concavity of Watch Time

To ensure a finite log-likelihood (LL), it is essential to ensure that the model can generate all possible actions. Specifically, the cost shock draws γ_{jt} must result in all streaming hour choices being selected with positive probability. For this, it is very helpful to impose that the marginal revenue from increasing streaming hours is non-increasing.

If this condition is not met, as depicted in Figure B.10, certain hour choices will be strictly dominated by other options and will not be chosen in the model. Consequently, the model assigns zero probability to some observed hour choices, causing my log-likelihood to diverge to negative infinity.

B.4 Watch Time (XGBoost) Details

Model Selection I first investigate the best performing machine learning (ML) tool for making proper watch time predictions. To do so, I utilize `AutoGluon`, a library developed by Amazon that automates the application and comparison of multiple ML tools for a data set. With `AutoGluon`, I evaluate the out-of-sample predictive accuracy of various ML models, including tree-based models (Random Forest, Extra Trees, LightGBM, CatBoost, and XGBoost), neural network models, and K nearest neighbor. XGBoost was chosen after comparing their out-of-sample prediction performances.

Specifically, I compare the aforementioned models with four specifications: using the first or last month as a test set and using out-of-sample root mean squared error (RMSE) or mean absolute error (MAE) as a performance metric. For each specification, I compute the relative loss of each model. The relative loss is defined by how much worse a model's performance is compared to the model that showed the best out-of-sample performance. For example, if XGBoost shows the best performance of 10,000 out-of-sample MAE, and the NeuralNetTorch shows 11,000, the two models have a relative loss of 0.0 and 0.1, respectively, for the specification. These results appear in Table B.9.

Hyperparameters I determine hyperparameters with 7-fold cross validation as there are seven month data period. The chosen hyperparametes included: the number of estimators (trees) 20, max depth 15, learning rate (η) 0.1, minimum child weight 1, subsample ratio 0.8. $log(bookmark)$ turned out to be the most "important" determinant, from perspective of how many times a variable was used to split nodes, followed by streaming hour and ψ_c.

Shape and Smoothness Restrictions I additionally address two issues. First, XGBoost's raw watch time predictions may not exhibit concavity (with respect to streaming hours), which is essential for the reasons outlined in Appendix B.3.

Second, because XGboost uses step functions as its basis, watch time often remains flat even when independent variables change. This characteristic isn't problematic for independent variables like bookmarks, but it poses a challenge for the category-level weighted sum of streaming hours, denoted as ψ_{ct}. The issue arises in the outer loop of the full equilibrium search algorithm in section 5.1. In this algorithm, I search for equilibrium by adjusting ψ_{ct}. If watch time remains flat concerning ψ_{ct}, the algorithm

Table B.9. Comparison of machine learning models through the `AutoGluon` library.

Model	(1)	(2)	(3)	(4)	Mean Relative Loss
XGBoost	0.043	0.105	0.006	0.000	0.038
NeuralNetTorch	0.064	0.032	0.074	0.007	0.044
LightGBM	0.014	0.469	0.004	0.253	0.185
ExtraTreesMSE	0.017	0.481	0.007	0.283	0.197
CatBoost	0.000	0.507	0.009	0.308	0.206
LightGBMLarge	0.036	1.696	0.006	0.340	0.520
LightGBMXT	0.069	1.523	0.036	0.523	0.538
RandomForestMSE	0.125	1.569	0.012	0.526	0.558
NeuralNetFastAI	0.067	8.689	0.000	0.121	2.219
KNeighborsUnif	1.371	20.474	0.081	0.166	5.523
KNeighborsDist	1.371	20.474	0.081	0.166	5.523
Test set	First month		Last Month		
Performance Metric	RMSE	MAE	RMSE	MAE	

Notes: The numbers in the cells represent the relative loss, which is defined as the relative difference in out-of-sample RMSE/MAE compared to the best-performing model in each specification. The best model in each specification, therefore, has a relative loss of zero. In specification (2), there is no model with a loss of zero because the weighted ensemble model, not included in this table, performed the best.

may fail to converge. [3] To resolve these issues, I proceed as follows.

1. Train XGBoost model. The observation level is individual streamer-month. The dependent variable is watch time, and independent variables are bookmarks, streaming hours, broadcasting category, profitability (per watch time tipping revenue), and ψ_{ct}.[4] Note that, at this point, ψ_{ct} varies across months, and I impose watch time to decrease when ψ_{ct} increases.

2. Obtain watch time predictions at the state level while holding ψ_{ct} constant at the monthly average $\bar{\psi}_c$. Denote these watch time predictions as $\widehat{\mathcal{W}}(x, h, \bar{\psi}_c)$, where x and h represent a state, i.e., a combination of category, profitability, and bookmarks, and streaming hours, respectively.

[3]More specifically, without imposing some smoothness with respect to ψ_{ct}, I found that the algorithm often results in the following infinite loop: streamers initially believe that ψ_{ct} is small (indicating less competition), causing them to stream more, resulting in a realized ψ_{ct} that is large. Streamers then update their belief based on this realization, leading them to believe that ψ_{ct} is large (indicating more competition), and subsequently, they stream less, resulting in a realized ψ_{ct} that is small again. As a result, ψ_{ct} can oscillate between small and large values without converging.

[4]To be precise, all independent variables were discretized, and broadcasting category and profitability are individual streamer level variables.

3. To impose concavity with respect to streaming hours, begin by collecting watch time predictions at the $x, \bar{\psi}_c$ level, i.e., $\{\widehat{\mathcal{W}}(x, h = 0, \bar{\psi}_c), \widehat{\mathcal{W}}(x, h = 50, \bar{\psi}_c), \ldots, \widehat{\mathcal{W}}(x, h = 300, \bar{\psi}_c)\}$. Next, approximate these predictions using non-parametric concave functions.[5] Denote this approximated concave watch time by $\widehat{\mathcal{W}}^{conc}(x, h, \bar{\psi}_c)$.

4. Impose that watch time changes smoothly with respect to ψ_{ct}. To determine the degree of change, I utilize the across-month variation of predicted (raw) watch times, denoted as $\widehat{\mathcal{W}}(x, h, \psi_{ct})$. Specifically, for each state x, I estimate coefficients θ_0 and θ_1 in the following specification:

$$log\left(\widehat{\mathcal{W}}\left(x, h, \psi_{ct}\right)\right) - log\left(\widehat{\mathcal{W}}\left(x, h, \bar{\psi}_c\right)\right) = \theta_0 + \theta_1\left(\psi_{ct} - \bar{\psi}_c\right) \tag{7}$$

   θ_1 represents, at the state level, the extent to which watch time decreases (on average across streaming hours) when there is increased competition, denoted by ψ_{ct}. Because I imposed that predicted watch time should decrease with respect to ψ_{ct} during the training of XGBoost, θ_1 is negative for all states as expected. This is consistent with the assumption that a streamer's watch time decreases when other streamers in the same category collectively stream more.

5. Among the above outcomes, only $\widehat{\mathcal{W}}^{conc}(x, h, \bar{\psi}_c)$ is used during the estimation stage. This reflects the assumption that ψ_{ct} is fixed at observed average level and streamers have a rational expectation for it.[6]. When searching for the full equilibrium during the counterfactual stage, I use the adjusted version of watch time predictions, across ψ_{ct}, given by

$$\widehat{\mathcal{W}}^{adj}\left(x, h, \psi_{ct}\right) = exp\left(\hat{\theta}_0 + \hat{\theta}_1\left(\psi_{ct} - \bar{\psi}_c\right)\right) \cdot \widehat{\mathcal{W}}^{conc}\left(x, h, \bar{\psi}_c\right) \tag{8}$$

   where $\hat{\theta}_0, \hat{\theta}$ are from equation 7. $\widehat{\mathcal{W}}^{adj}$ maintains concavity with respect to streaming hours, and smoothly changes with respect to ψ_{ct}, as desired.

Watch Time Fit Assessment Watch time approximation errors may arise from three sources. First, prediction errors stem from the discretization of continuous

[5]For this approximation, I minimize the mean squared errors. The methodology was developed by Kuosmanen (2008) and Kuosmanen and Johnson (2010). I implement their method using python `cvxopt` package and `mosek` optimization tool.

[6]From this perspective, I am assuming that all data points come from one equilibrium, while the endogenous variable ψ_{ct}, which differs across months, is represented as $\bar{\psi}_c + e_{ct}$. I am disregarding e_{ct} during the estimation stage because ψ_{ct} does not vary significantly across months

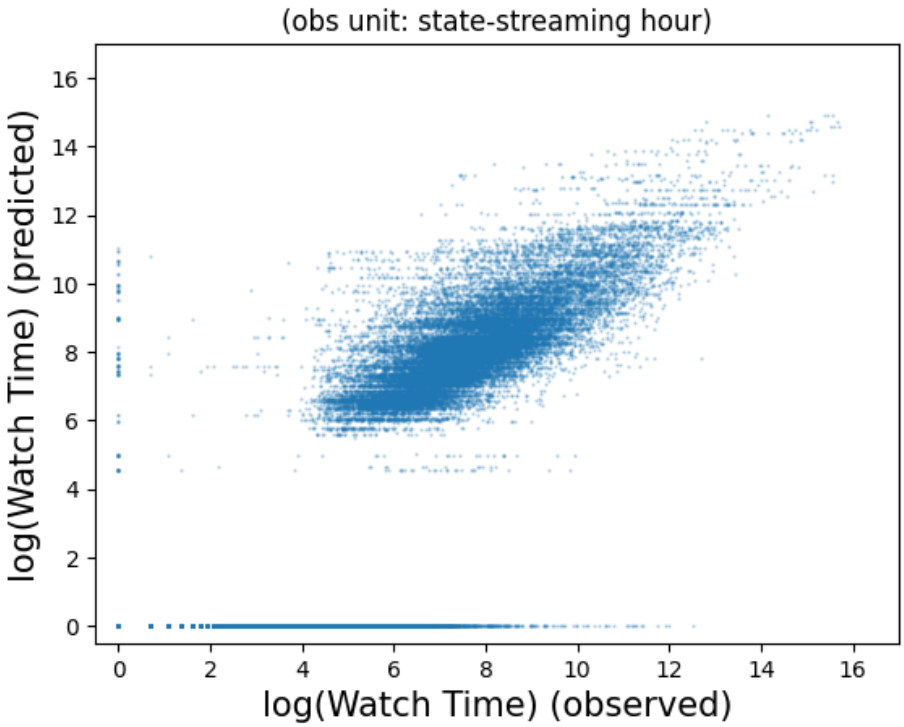


Figure B.11. Observed vs. predicted watch time (specifically, $\widehat{\mathcal{W}}^{conc}(x, h, \bar{\psi}_c)$, i.e., predictions with concavity imposed). The unit of observation is a state–action (streaming hour) pair. Further discussion of the horizontal blue band at the bottom is provided in the paragraph below.

variables. Second, additional approximation errors arise from the ML model and the imposition of concavity on watch time with respect to streaming hours. Lastly, I assume that the weighted sum of hours remains fixed at its monthly mean, whereas in reality it may vary across months.

To assess how well the XGBoost model with shape restrictions approximates watch time, I plot observed versus predicted watch time (specifically, $\widehat{\mathcal{W}}^{conc}(x, h, \bar{\psi}_c)$, i.e., predictions with concavity imposed) at the state–action (streaming hour) level (Figure B.11).

One notable feature is the horizontal blue band at the bottom, which suggests that some observations with positive watch time are predicted to be zero. However, further investigation indicates that this pattern is not a major concern: the average observed watch time for observations with zero predicted watch time is 312.23, whereas the overall average observed watch time is 11,433.26. The log transformation used for visualization amplifies discrepancies at low values.

Two factors contribute to this pattern. First, the imposition of concavity with

respect to streaming hours, together with predicted watch time at zero hours being close to zero, tends to generate such outcomes. Second, because streaming hours are discretized into $\{0, 50, \ldots, 300\}$, some streamers who broadcast for around 10–20 hours are treated as having zero hours, even though their observed watch time is not negligible.

C Additional Model Fit Assessments

C.1 Predicted vs. Observed Industry state

Using the full equilibrium search algorithm, I additionally investigate if the distribution of score and bookmarks at industry level from factual simulation are close to observed counterpart. This counterpart was taken as given throughout the estimation stage. The assessment appears in Figure C.12.

Overall, the largest gaps come from the fraction of highly popular streamers, and it may have come from potential bias in entry process. A streamer tracking website started tracking on January 2019, and my data period starts from October 2019 since revenue information is available from then.

It could be the case that the website started to track (relatively) popular streamers first and less popular streamers later. In this case, in the entry process I am using, less popular streamers may have been overrepresented.

Additionally, I compare various industry-level state variables from factual simulations with the observed averages, as shown in Table C.10. The main takeaway is that the predictions from factual simulations, while not perfect, are not too far from the observed industry state. This finding is encouraging, as the estimation process using pseudo-MLE did not directly target matching these features.

Figure C.12. Observed vs. predicted marginal distributions of *score* and *log*(*bookmark*)

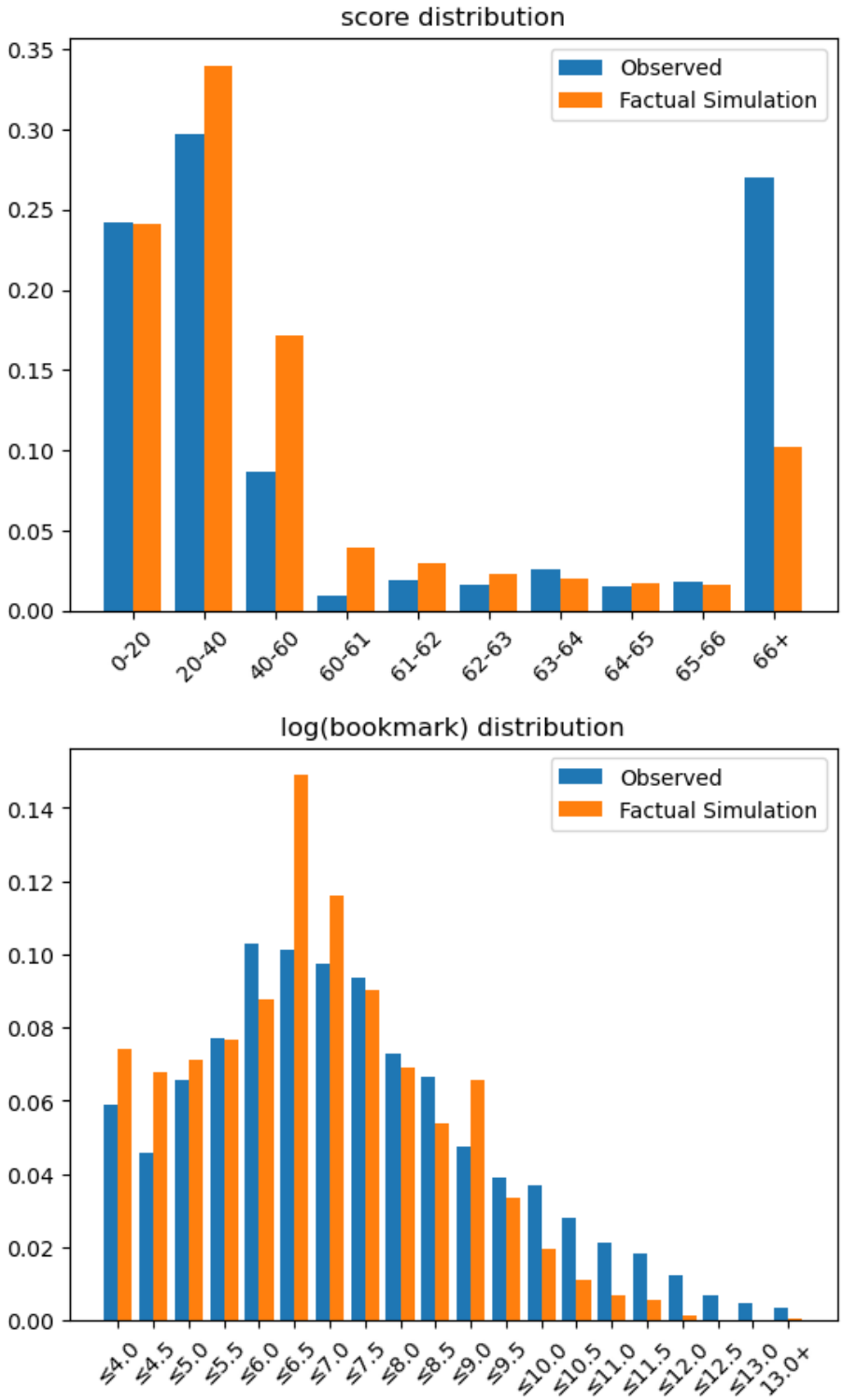


Table C.10. Predicted versus observed platform level outcomes.

	Observed	Predicted
Streamers newly getting the reward		
Total number	38.00	21.36
Game (category share)	0.39	0.57
Social	0.36	0.21
Other	0.25	0.21
Low (α^{rev} share)	0.14	0.15
Mid	0.31	0.33
High	0.56	0.51
Weighted sum of streaming hour ψ_{ct} (Unit: 1,000)		
Game (category)	1827.13	2397.42
Social	725.69	812.12
Other	962.43	949.58
The number of streamers	5888.71	8034.73
Total revenue (Unit: 1,000 USD)	188.91	229.47
Platform's revenue share	0.34	0.35

Notes: The numbers in the left column are from data averaged across seven months. The numbers in the right column are from factual simulation endogenizing ψ_{ct}. The shares may not add up to one due to rounding errors.

C.2 Out-of-sample assessment: doubling-slot counterfactual

Counterfactual 1 (doubling slots) yields a prediction for the new approval probabilities by score when the number of slots is doubled. In practice, this change was implemented, allowing me to compare the prediction with realized data and conduct an out-of-sample evaluation of model fit.

First, I plot empirical approval probabilities and cutoffs for two periods: the main data period (October 2019 to April 2020) and the period after the slot increase, once cutoffs had stabilized (August 2020 to October 2021). I use 2-point score bins, with the x-axis representing the average score within each bin. (The decline in approval probabilities well above the cutoff may reflect the presence of popular streamers who are consistently uninterested in the rewards program; I discuss this as a limitation below.)

The main takeaway is that the average score cutoff decreased from 65.52 to 60.68, which is broadly consistent with the counterfactual prediction. As shown in Figure C.14, counterfactual simulation 1 predicts that, after doubling the number of slots and reaching a steady state, all streamers with scores above 61 are always approved (conditional on applying), while those with scores between 60 and 61 are approved with positive probability.

Finally, one confounding factor remains. In practice, there was a minor change in the score calculation when the number of slots increased: streamers with an average viewer count below 10 began to receive 0.4×50 points instead of zero. To ensure comparability across periods, I recompute all scores using the original method. Because the change is minor, the two measures are highly correlated and have similar magnitudes.

C.3 Deviation Factors

Several factors may contribute to the gap between the model's predictions and the observed data, as listed below. I do not repeat factors discussed in Section 3.2 (e.g., the presence of streamers who are persistently uninterested in the rewards program).

First, I discretize continuous variables. For example, I find that streamers who are accepted into the rewards program tend to have higher per-watch-time revenue parameters, α^{rev}, within a given discretized group. In the Social–low α^{rev} group, the discretized value is 0.239, whereas newly approved streamers in this group have an average α^{rev} of 0.488.

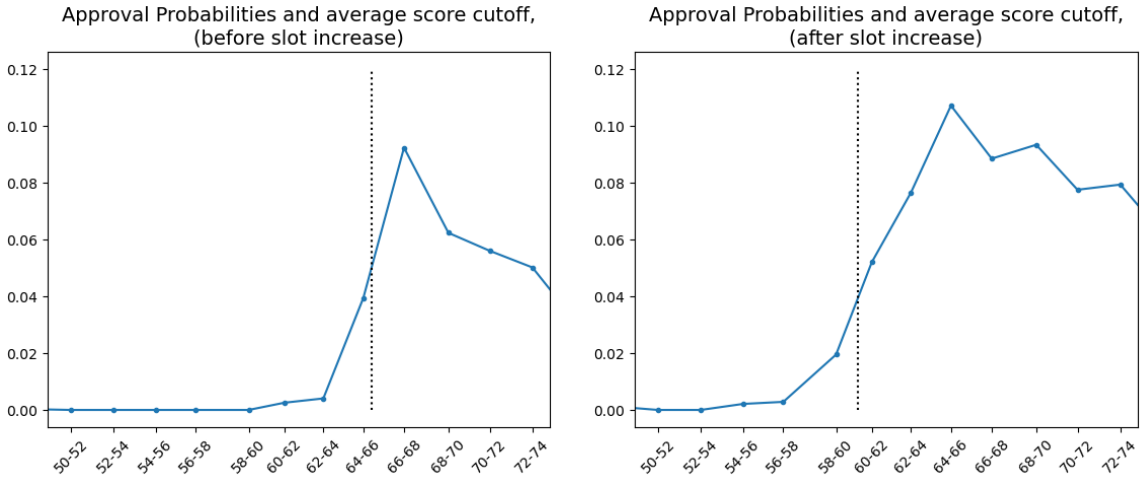


Figure C.13. Observed approval probabilities by score and average cutoff. Left: October 2019 to April 2020. Right: August 2020 to October 2021.

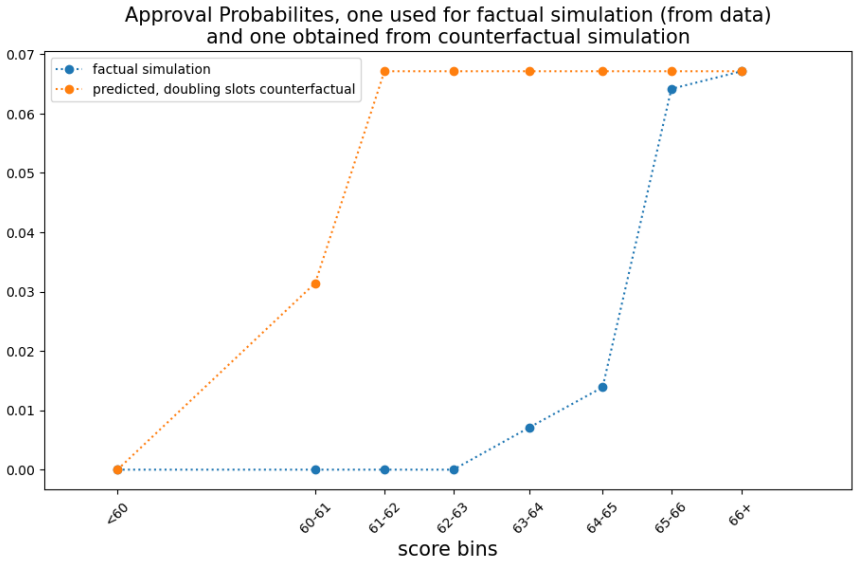


Figure C.14. Change in approval probabilities from counterfactual simulation 1

Second, I use the entry process observed during the data period. However, a substantial fraction of streamers in the sample likely entered the market well before the data period, possibly under a different entry process.

Third, the equilibrium of the dynamic game described in Section 3.1.3 is assumed to approximate streamers' decisions. However, Table C.10 shows a gap between the endogenized (predicted) and observed weighted sum of streaming hours. This suggests that the observed data may not fully align with the equilibrium concept.